Chemistry Grade 11 - Solved Exercises (BTBB)

Balochistan Textbook Board (BTBB)

Unit 1: Scientific Thinking and Stoichiometry

Exercise - Multiple Choice Questions (MCQs)

Q(i). A thought experiment is:

- A physical experiment
- A real laboratory test
- A mathematical expression
- ✓ A hypothetical scenario involving scientific ideas

Q(ii). The branch of chemistry that describes the relationship between the amounts of reactants and products in a balanced chemical equation is called:

- Physical chemistry
- Biochemistry
- ✓ Stoichiometry
- Organic chemistry

Q(iii). The number of covalent bonds in 68 g of H2S gas is:

- 3.01 x 10^23
- 6.02 x 10^23
- ✓ 2.41 x 10^24
- 24.1 x 10^24

Q(iv). The mass of O2 required to burn 0.1 mole of C2H5OH is: (Equation: C2H5OH + 3O2 -> 2CO2 + 3H2O)

- 32 g

3.2 g

5.6 g

✓ 9.6 g

Q(v). The volume occupied by 1.4 g of N2 at STP is:

2.24 dm^3

22.4 dm^3

✓ 1.12 dm^3

112 dm^3

Q(vi). A beaker contains 9 g of water. The number of hydrogen atoms is:

✓ 6.02 x 10^23

3.01 x 10^23

6.02 x 10^24

3.01 x 10^24

Q(vii). One mole of a diamond chain and one mole of a gold ring have the same number of:

Neutrons

Protons

Electrons

✓ Atoms

Q(viii). The largest number of molecules is present in:

4.8 g of C2H5OH

✓ 3.6 g of H2O

2.8 g of CO

5.4 g of N2

Q(ix). The amount of product obtained practically is called:

Expected yield

Theoretical yield

✓ Actual yield

Fractional yield

Q(x). The reactant that is in a larger amount and remains unreacted is called:

Limiting reactant

✓ Excess reactant

Expensive reactant

Restricting the reactant

Exercise - Restricted Response Questions (RRQs)

Q(i). Why are thought experiments essential in chemical theory?

Ans. Thought experiments are essential in chemical theory because they let chemists test a hypothesis, theory or principle mentally, without doing a physical experiment. They act as a 'mental laboratory': by imagining a scenario (for example, equal volumes of gases at the same temperature and pressure) and following its logical consequences, chemists can predict outcomes, expose errors in a theory, and understand abstract ideas such as Avogadro's hypothesis and the mole, which cannot easily be seen or handled directly.

Q(ii). Why is stoichiometry important?

Ans. Stoichiometry is important because it is the quantitative side of chemistry: from a balanced equation it tells us exactly how much of each reactant is needed and how much product will form. Industry uses it to calculate raw materials, costs and yields; laboratories use it to prepare solutions of exact concentration; and it underlies the ideas of limiting reactants, percentage yield and gas volumes, without which chemical manufacture would be guesswork.

Q(iii). What is the difference between a chemical reaction and a chemical equation?

Ans. A chemical reaction is the actual process in which reactants change into products, with bonds breaking and forming (for example, iron and sulphur heating to form iron sulphide). A chemical equation is the written, symbolic representation of that reaction, using formulae and symbols (for example, Fe + S -> FeS), balanced so that the number of atoms of each element is the same on both sides, as required by the law of conservation of mass.

Q(iv). Why do Avogadro's number of atoms of different elements have different masses?

Ans. Avogadro's number is the same for every element — one mole of any element contains 6.02 x 10^23 atoms — but the atoms of different elements have different masses, because each element's atoms have a different number of protons and neutrons. So one mole of sodium atoms weighs 23 g while one mole of calcium atoms weighs 40 g, even though both contain the same number of atoms; the mass differs because the individual atoms differ in

mass.

Q(v). Which would have a higher mass: a mole of Na atoms or a mole of Ca atoms?

Ans. A mole of calcium atoms has the higher mass. One mole of sodium atoms weighs 23 g (atomic mass of Na = 23), while one mole of calcium atoms weighs 40 g (atomic mass of Ca = 40). Both contain 6.02 x 10^23 atoms, but calcium atoms are heavier, so the mole of calcium is heavier.

Q(vi). Why is the actual yield of a reaction always smaller than the theoretical yield?

Ans. The actual yield is always smaller than the theoretical yield because reactions rarely go to completion; some reactant remains unreacted; side reactions produce unwanted products; some product is lost during separation, filtration, transfer and purification; and impure reactants mean less pure reactant is actually available than weighed out. The theoretical yield assumes a perfect, complete reaction with no losses, which never happens in practice.

Exercise - Extended Response Questions (ERQs)

Q(i). Define and explain mole and Avogadro's number with examples.

Ans. A mole is the amount of substance that contains as many particles (atoms, molecules or ions) as there are atoms in exactly 12 g of carbon-12. That number is Avogadro's number, 6.02 x 10^23. So one mole of hydrogen atoms weighs 1 g, one mole of

oxygen molecules weighs 32 g, and one mole of water weighs 18 g — in every case the sample contains 6.02 x 10^23 particles. Examples: 12 g of carbon, 23 g of sodium and 40 g of calcium each contain one mole of atoms; 18 g of water contains one mole of water molecules (and two moles of hydrogen atoms).

Q(ii). Define and explain molar mass and molar volume with examples. How can we calculate the molar mass of a substance?
Ans. Molar mass is the mass of one mole of a substance, in grams per mole; it is numerically equal to the atomic or molecular mass. Examples: molar mass of H2O = 18 g/mol, of NaCl = 58.5 g/mol, of CO2 = 44 g/mol. Molar volume is the volume occupied by one mole of an ideal gas at STP, which is 22.414 dm^3 (22.4 dm^3). Examples: one mole of hydrogen, oxygen or any gas occupies 22.4 dm^3 at STP. To calculate the molar mass of a substance, add the atomic masses of all atoms in its formula: for example, H2SO4 = (2 x 1) + 32 + (4 x 16) = 98 g/mol.

Q(iii). What are limiting and excess reagents? How will you determine them?
Ans. The limiting reactant (limiting reagent) is the reactant that is used up first and limits the amount of product formed; the excess reactant is the one left over when the reaction ends. To determine them: (1) write the balanced equation; (2) convert the given masses (or volumes) of reactants to moles; (3) divide each mole value by its coefficient in the equation; (4) the reactant with the smallest value is the limiting reactant, and the others are in excess. Example: in 2H2 + O2 -> 2H2O, if hydrogen runs out first,

hydrogen is limiting and oxygen is in excess.

Q(iv). Define theoretical, actual yield, and percentage yield. How do we calculate the percentage yield of a chemical reaction?

Ans. Theoretical yield is the maximum amount of product calculated from the balanced equation, assuming complete reaction. Actual yield is the amount actually obtained in practice. Percentage yield = (actual yield / theoretical yield) x 100. To calculate it: (1) find the limiting reactant and the theoretical yield from the balanced equation; (2) weigh the actual product obtained; (3) divide actual by theoretical and multiply by 100. Example: if the theoretical yield of FeS is 44.07 kg and 40 kg is obtained, percentage yield = (40 / 44.07) x 100 = 90.76%.

Exercise - Numerical Problems

Q(i). How much iron is required to produce 162.3 g of FeCl3, when chlorine is in excess? 2Fe(s) + 3Cl2(g) -> 2FeCl3(s).

Step 1: Molar mass of FeCl3 = 55.85 + (3 x 35.5) = 162.35 g/mol (approx. 162.3 g/mol).

Step 2: Moles of FeCl3 required = 162.3 g / 162.3 g/mol = 1.00 mol.

Step 3: From the equation, 2 mol Fe produce 2 mol FeCl3, so moles of Fe needed = 1.00 mol.

Step 4: Mass of Fe = 1.00 mol x 55.85 g/mol = 55.85 g (book answer: 55.89 g).

Answer: 55.89 g of iron.

Q(ii). Calculate how many moles of oxygen are required to make 270 g of aluminum oxide. The balanced equation is: 4Al(s) + 3O2(g) -> 2Al2O3(s).

Step 1: Molar mass of Al2O3 = (2 x 27) + (3 x 16) = 102 g/mol.

Step 2: Moles of Al2O3 = 270 g / 102 g/mol = 2.647 mol.

Step 3: From the equation, 2 mol Al2O3 need 3 mol O2, so moles of O2 = 2.647 x (3/2) = 3.97 mol.

Answer: 3.97 moles of O2.

Q(iii). How many moles are in 25 g of each of the following substances? (a) F2 (Fluorine) (b) Ni (Nickel) (c) CS2 (Carbon disulphide) (d) C2F2Cl4 (Freon gas) (e) FeSO4 (Ferrous sulphate).

Step 1: (a) Molar mass of F2 = 38 g/mol; moles = 25 / 38 = 0.658 mol.

Step 2: (b) Molar mass of Ni = 58.7 g/mol; moles = 25 / 58.7 = 0.426 mol.

Step 3: (c) Molar mass of CS2 = 12 + (2 x 32) = 76 g/mol; moles = 25 / 76 = 0.328 mol.

Step 4: (d) Molar mass of C2F2Cl4 = (2 x 12) + (2 x 19) + (4 x 35.5) = 204 g/mol; moles = 25 / 204 = 0.123 mol.

Step 5: (e) Molar mass of $FeSO_4$ = 56 + 32 + (4 x 16) = 152 g/mol; moles = 25 / 152 = 0.165 mol.

Answer: (a) 0.658 mol, (b) 0.426 mol, (c) 0.328 mol, (d) 0.123 mol, (e) 0.165 mol.

Q(iv). How many particles are present in 2.5 moles of carbonate ion and in 1 mole of aspirin ($C_9H_8O_4$)?

Step 1: Particles = moles x Avogadro's number (6.022 x 10^23).

Step 2: Carbonate ions = 2.5 x 6.022 x 10^23 = 1.506 x 10^24 ions.

Step 3: Aspirin molecules = 1 x 6.022 x 10^23 = 6.022 x 10^23 molecules.

Answer: 1.506 x 10^24 carbonate ions and 6.022 x 10^23 aspirin molecules.

Q(v). What is the volume of 1.25 moles of SO_2 and 0.50 moles of argon gas at STP?

Step 1: Total moles of gas = 1.25 + 0.50 = 1.75 mol.

Step 2: At STP, 1 mol of any gas occupies 22.4 dm^3.

Step 3: Volume = 1.75 mol x 22.4 dm^3/mol = 39.2 dm^3 (39.2 litres).

Answer: 39.2 litres.

Q(vi). How much water can the reaction make from 50 g of hydrogen and 50 g of oxygen? 2H2(g) + O2(g) -> 2H2O(g).
Step 1: Moles of H2 = 50 g / 2 g/mol = 25 mol; moles of O2 = 50 g / 32 g/mol = 1.5625 mol.

Step 2: From the equation, 1 mol O2 reacts with 2 mol H2, so 1.5625 mol O2 needs 3.125 mol H2. Hydrogen (25 mol) is in excess; oxygen is the limiting reactant.

Step 3: Moles of H2O formed = 2 x 1.5625 = 3.125 mol.

Step 4: Mass of H2O = 3.125 mol x 18.02 g/mol = 56.31 g.

Answer: 56.31 g of water.

Q(vii). 4.64 g of iron reacts completely with an excess of CuSO4 and gives 5 g of copper. Fe(s) + CuSO4(aq) -> FeSO4(aq) + Cu(s). (a) What is the theoretical yield of the reaction? (b) What is the percentage yield of the reaction?
Step 1: Moles of Fe = 4.64 g / 55.85 g/mol = 0.0831 mol.

Step 2: From the equation, 1 mol Fe produces 1 mol Cu, so moles of Cu = 0.0831 mol.

Step 3: Theoretical yield of Cu = 0.0831 mol x 63.5 g/mol = 5.28 g.

Step 4: Percentage yield = (actual yield / theoretical yield) x 100 = (5 g / 5.28 g) x 100 = 94.70%.

Answer: (a) Theoretical yield of copper = 5.28 g. (b) Percentage yield of copper = 94.70%.

Self-Assessment (model answer)

Iron sulphide is produced by heating iron with sulphur: Fe(s) + S(s) -> FeS(s). When 28 kg of iron is combined with excess of sulphur, 40 kg of iron sulphide (FeS) is formed. Calculate the percentage yield of the FeS.

Step 1: Moles of Fe = 28,000 g / 55.85 g/mol = 501.3 mol.

Step 2: From the equation, 1 mol Fe produces 1 mol FeS, so theoretical moles of FeS = 501.3 mol.

Step 3: Molar mass of FeS = 55.85 + 32.06 = 87.91 g/mol; theoretical yield = 501.3 x 87.91 = 44,070 g = 44.07 kg.

Step 4: Percentage yield = (40 kg / 44.07 kg) x 100 = 90.76%.

Answer: Percentage yield = 90.76%.

Unit 2: Atomic Structure

Exercise - Multiple Choice Questions (MCQs)

Q(i). The fundamental particles of an atom are:

- Electron, positron, neutron
- ✓ Electron, proton, neutron
- Electron, neutrino, proton
- Electron, positron, meson

Q(ii). Proton was discovered by:

- ✓ Rutherford
- Chadwick
- Thomson
- W. Crooks

Q(iii). Select the atom that has a half-filled 2p subshell:

- Carbon
- ✓ Nitrogen
- Oxygen
- Fluorine

Q(iv). The mass of an electron is almost equal to:

- Hydrogen atom
- Proton
- Neutron
- ✓ Positron

Q(v). The atomic radius of group-IA elements:

- Decrease down the group
- ✓ Increase down the group
- Remains the same in the group
- First decreases and then increases

Q(vi). Mass spectrometer separates ions on the basis of:

- Mass
- Charge
- ✓ Mass to charge ratio
- Molecular Mass

Q(vii). Identify the principle that states no two electrons in an atom can have the same four quantum numbers.

- Hund's Rule
- Aufbau Principle
- ✓ Pauli Exclusion Principle
- Heisenberg Principle

Q(viii). When the 6s orbital is complete, the entering electron goes into:

- 7s
- 6p
- 5d
- ✓ 4f

Q(ix). If n is equal to 4, then l is equal to:

0

0, 1

0, 1, 2

✓ 0, 1, 2, 3

Q(x). The maximum number of electrons an orbital can accommodate:

14

10

6

✓ 2

Exercise - Restricted Response Questions (RRQs)

Q(i). Explain the behaviour of protons, electrons, and neutrons when moving in an electric field at the same velocity.

Ans. In an electric field, a proton (positively charged) is deflected towards the negative plate, an electron (negatively charged) is deflected towards the positive plate, and a neutron (uncharged) passes straight through undeflected. The electron is deflected much more than the proton because it is far lighter, even though both carry equal and opposite charges.

Q(ii). What is the total number of subshells and orbitals in an atom whose principal quantum number n = 4?

Ans. For n = 4, the subshells are 4s, 4p, 4d and 4f — a total of 4 subshells. The total number of orbitals is n^2 = 16 (1 in 4s, 3 in 4p, 5 in 4d and 7 in 4f), holding a maximum of 32 electrons.

Q(iii). How many orbitals are in the 2p subshell?

Ans. There are 3 orbitals in the 2p subshell (2px, 2py, 2pz), each holding a maximum of 2 electrons, so the 2p subshell holds 6 electrons in total.

Q(iv). Which subatomic particle contributes to the identity of an element?

Ans. The proton contributes to the identity of an element: the number of protons (the atomic number, Z) is unique to each element and identifies it. All atoms of one element have the same number of protons; changing the number of protons changes the element itself.

Q(v). State the formula used to calculate the total number of orbitals in a subshell.

Ans. The number of orbitals in a subshell is given by (2l + 1), where l is the azimuthal quantum number. So an s subshell (l = 0) has 1 orbital, a p subshell (l = 1) has 3, a d subshell (l = 2) has 5, and an f subshell (l = 3) has 7 orbitals.

Q(vi). Which subshell in the following pairs is higher in energy? (i) 4p or 4d (ii) 5s or 6p (iii) 6s or 4f.

Ans. Using the (n + l) rule, the subshell with the higher (n + l) value has higher energy. (i) 4p (n + l = 5) and 4d (n + l = 6): 4d is higher in energy. (ii) 5s (n + l = 5) and 6p (n + l = 7): 6p is higher in energy. (iii) 6s (n + l = 6) and 4f (n + l = 7): 4f is higher in energy.

Exercise - Extended Response Questions (ERQs)

Q(i). Describe the shapes of s, p, and d-orbitals. How are these orbitals related to azimuthal and magnetic quantum numbers?

Ans. The s-orbital is spherical, with the electron density highest near the nucleus and falling off in all directions; it is described by l = 0 and has only one orientation (m = 0). The p-orbitals are dumb-bell shaped, with two lobes on opposite sides of the nucleus; they occur for l = 1 and have three orientations (m = -1, 0, +1), called px, py and pz. The d-orbitals are mostly clover-leaf shaped (four lobes), with one (dz^2) having a doughnut shape; they occur for l = 2 and have five orientations (m = -2, -1, 0, +1, +2). Thus the azimuthal quantum number l decides the shape of the orbital, and the magnetic quantum number m decides its orientation in space.

Q(ii). Discuss the rules governing the distribution of electrons in subshells (Aufbau Principle, Pauli Exclusion Principle, Hund's Rule, and their application).

Ans. The distribution of electrons in subshells is governed by three rules. (1) Aufbau principle: electrons fill subshells in order of increasing energy, following the (n + l) rule — for example 1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p. (2) Pauli Exclusion Principle: no two electrons in an atom can have all four quantum numbers the same, so an orbital holds at most two electrons, with opposite spins. (3) Hund's rule: in degenerate orbitals (same energy, such as the three p orbitals), electrons occupy separate orbitals singly first, with parallel spins, and pairing begins only after each orbital has one electron. Application: nitrogen (1s^2 2s^2 2p^3) has three unpaired electrons in the three 2p orbitals; oxygen (1s^2 2s^2 2p^4) has one pair and two unpaired electrons.

Q(iii). Discuss the differences between atomic radius and ionic radius. Explain why cations are smaller and anions are larger compared to their parent atoms.

Ans. Atomic radius is half the distance between the nuclei of two bonded atoms of the same element, while ionic radius is the radius of an ion in a crystal. Cations are smaller than their parent atoms because the atom loses one or more outer electrons: the remaining electrons are pulled closer by the same nuclear charge, and sometimes a whole shell is lost (for example Na, 186 pm, becomes Na+, 102 pm). Anions are larger than their parent atoms because gaining electrons increases electron-electron repulsion and reduces the effective nuclear attraction per electron, so the electron cloud expands (for example Cl, 99 pm, becomes Cl-, 181 pm).

Self-Assessment (model answer)

Q1. Use the given ionisation data for aluminium to identify when a core electron is removed. What does this tell us about its position in the Periodic Table?

Ans. Aluminium's successive ionisation energies are 577, 1817, 2745 and 11577 kJ/mol. The first three increases are small, but the fourth is a huge jump. This shows that the first three electrons are removed from the outer shell (valence electrons), while the fourth electron comes from a stable inner (core) shell, which is held much more tightly. So aluminium has 3 valence electrons, placing it in Group 13 (Group IIIA) and, since the valence shell is the third shell, in Period 3 of the periodic table.

Unit 3: Chemical Bonding

Exercise - Multiple Choice Questions (MCQs)

Q(i). The first bond formed between two atoms is:

- ✓ Sigma bond
- Pi-bond
- Double bond
- Triple bond

Q(ii). Molecules with four bond pairs and no lone pairs have the following geometry:

- Trigonal
- ✓ Tetrahedral
- Octahedral
- Triangular

Q(iii). The compound with a linear shape is:

- ✓ CO2
- SO2
- O3
- H2O

Q(iv). Indicate the molecule that shows sp^3 hybridisation:

- ✓ NH3
- AlCl3
- MgCl2

BeH2

Q(v). The example of polar molecules is:

H2

CH2

CCl4

✓ CHCl3

Q(vi). The s-character in each sp^2 hybrid orbital is:

25%

✓ 33%

50%

75%

Q(vii). The number of lone pairs of electrons in NH3 is:

4

3

2

✓ 1

Q(viii). The second most electronegative atom is:

Nitrogen

Fluorine

Chlorine

✓ Oxygen

Q(ix). The example of a paramagnetic molecule is:

H2

N2

✓ O2

F2

Q(x). The acetylene molecule has:

One sigma and one pi-bond

✓ One sigma and two pi-bonds

Two sigma and two pi-bonds

Three sigma and two pi-bonds

Exercise - Restricted Response Questions (RRQs)

Q(i). Why does the magnitude of repulsion increase in the following order: bond pair-bond pair < bond pair-lone pair < lone pair-lone pair?

Ans. Lone pairs are held by only one nucleus, so they spread out more and occupy more space around the central atom than bond pairs, which are held between two nuclei. Therefore a lone pair repels neighbouring pairs more strongly than a bond pair does, giving the order: bond pair-bond pair repulsion < bond pair-lone pair repulsion < lone pair-lone pair repulsion. This is why, for example, the H-N-H angle in ammonia (107 degrees) is less than the tetrahedral angle (109.5 degrees).

Q(ii). How many atoms are directly bonded to the central atom in a linear, tetrahedral, and octahedral molecule?

Ans. In a linear molecule (for example CO2, BeCl2), 2 atoms are directly bonded to the central atom. In a tetrahedral molecule (for example CH4, NH4+), 4 atoms are directly bonded to the central

atom. In an octahedral molecule (for example SF6, PCl6-), 6 atoms are directly bonded to the central atom.

Q(iii). Do all the sigma molecular orbitals result from the overlap of s atomic orbitals?

Ans. No. Sigma molecular orbitals result from the head-on (linear) overlap of atomic orbitals, which may be s-s, s-p or p-p overlap. Only the s-s combination involves two s orbitals; for example, the sigma bond in H2 comes from s-s overlap, in HCl from s-p overlap, and in Cl2 from p-p overlap. So sigma molecular orbitals arise from several kinds of overlap, not only from s orbitals.

Q(iv). Why are polar covalent bonds stronger than nonpolar covalent bonds?

Ans. In a polar covalent bond the shared electrons are pulled towards the more electronegative atom, giving partial charges (delta+ and delta-) at the ends. This charge separation adds an electrostatic (ionic) attraction to the covalent sharing, making the bond stronger and shorter than a comparable non-polar bond, in which the electrons are shared equally and no extra ionic character is present. Generally, the greater the electronegativity difference, the greater the partial ionic character and the stronger the bond.

Q(v). Separate those molecules from the molecules given below which have a dipole moment equal to zero: (i) HBr (ii) Cl2 (iii) SO2 (iv) C2H2 (v) H2O (vi) CS2 (vii) C6H6 (viii) CHCl3 (ix) NH3 (x) SiH4.

Ans. Zero dipole moment (non-polar, symmetrical molecules): (ii) Cl2, (iv) C2H2, (vi) CS2, (vii) C6H6, (x) SiH4 — in these, the

individual bond dipoles cancel because of symmetry. Non-zero dipole moment (polar molecules): (i) HBr, (iii) SO2, (v) H2O, (viii) CHCl3, (ix) NH3 — their bond dipoles do not cancel, either because the bonds are between different atoms or because the shape is unsymmetrical.

Exercise - Extended Response Questions (ERQs)

Q(i). Describe the key characteristics of valence bond theory and outline its disadvantages.

Ans. Valence bond theory says a covalent bond forms when two half-filled atomic orbitals overlap, and the shared pair of electrons occupies the overlapped region, holding the two nuclei together. Key characteristics: bonds form by orbital overlap (head-on overlap gives sigma bonds, side-by-side overlap gives pi bonds); the strength of a bond depends on the extent of overlap; and the shapes of molecules are explained by hybridisation (sp^3 tetrahedral, sp^2 trigonal planar, sp linear). Disadvantages: it cannot explain the paramagnetic character of O2 (which has unpaired electrons); it does not explain the bonding in delocalised systems such as benzene satisfactorily; and it gives no quantitative account of bond energies or spectra.

Q(ii). What does hybridisation of atomic orbitals mean? Discuss the different types of hybridisation.

Ans. Hybridisation is the mixing of atomic orbitals of similar energy in an atom to form an equal number of new, equivalent hybrid orbitals, which then form bonds of equal strength and in definite directions. Types: sp hybridisation — one s and one p orbital mix to

give two sp orbitals, linear geometry, 180 degrees (for example BeCl2, C2H2); sp^2 hybridisation — one s and two p orbitals mix to give three sp^2 orbitals, trigonal planar geometry, 120 degrees (for example BF3, C2H4); sp^3 hybridisation — one s and three p orbitals mix to give four sp^3 orbitals, tetrahedral geometry, 109.5 degrees (for example CH4, NH3).

Q(iii). What are the main points of molecular orbital theory, and how does it differ from valence bond theory?

Ans. Main points of molecular orbital theory: when atoms combine, their atomic orbitals mix to form molecular orbitals that belong to the whole molecule — bonding molecular orbitals (lower energy, formed by constructive overlap) and antibonding molecular orbitals (higher energy, formed by destructive overlap, marked with *). Electrons fill these in order of increasing energy, following Aufbau, Pauli and Hund rules; bond order = (bonding electrons - antibonding electrons) / 2. Difference from valence bond theory: valence bond theory keeps electrons localised between two atoms as shared pairs, while molecular orbital theory delocalises electrons over the whole molecule. Molecular orbital theory therefore explains the paramagnetism of O2 (two unpaired electrons in antibonding pi orbitals) and the existence of species such as H2+ and He2, which valence bond theory cannot.

Q(iv). Define intramolecular and intermolecular forces and list the various types of each.

Ans. Intramolecular forces act within a molecule, holding its atoms together: ionic bonds (electrostatic attraction between oppositely

charged ions), covalent bonds (sharing of electron pairs) and metallic bonds (attraction between metal cations and the sea of delocalised electrons). Intermolecular forces act between separate molecules and are much weaker: dipole-dipole forces (between polar molecules), London dispersion forces (temporary dipoles in all molecules), hydrogen bonding (a strong dipole interaction when hydrogen is bonded to F, O or N), and ion-dipole forces (between an ion and a polar molecule). Intramolecular forces decide a compound's chemical identity; intermolecular forces decide its melting point, boiling point and solubility.

Self-Assessment (model answers)

Q1. Ammonia reacts with hydrogen chloride to form ammonium chloride. Write the balanced equation. Explain dative bonding in the ammonium ion and the bonding type in solid ammonium chloride.

Ans. Ammonia reacts with hydrogen chloride as: NH3(g) + HCl(g) -> NH4Cl(s). In the ammonium ion (NH4+), the nitrogen atom of NH3 donates its lone pair to the hydrogen ion (H+) from HCl, forming a dative (coordinate) bond: the nitrogen shares its lone pair with H+, so all four N-H bonds become equivalent. In solid ammonium chloride, the NH4+ ions and Cl- ions are held together by ionic bonding (electrostatic attraction), so the solid is an ionic compound even though the ammonium ion itself contains covalent and dative bonds.

Q2. Why is CO2 linear and water angular? Justify your answer using the concept of dipole moment.

Ans. CO2 is linear because the carbon atom has two bond pairs and

no lone pairs, so the two C=O bond dipoles point in exactly opposite directions and cancel, giving zero dipole moment. Water is angular (bent) because the oxygen atom has two bond pairs and two lone pairs; the lone pairs push the O-H bonds down into a bent shape (about 104.5 degrees), so the two O-H bond dipoles do not cancel and the molecule has a net dipole moment of 1.85 D. Thus CO2 is non-polar despite polar bonds, while water is polar.

Unit 4: States of Matter

Exercise - Multiple Choice Questions (MCQs)

Q(i). The gases are compressible because:

- Their molecules are heavy
- Their molecules are closely packed
- ✓ There is significant empty space between gas molecules
- Their intermolecular forces are negligible

Q(ii). Glycerin is more viscous than water because:

- It has a lower density
- It is non-polar
- ✓ It forms more hydrogen bonds
- It has weaker cohesive forces

Q(iii). The liquid with weak intermolecular forces is:

- Diesel
- Water
- Ethyl alcohol
- ✓ Petrol

Q(iv). The conversion of vapours into liquid is called:

- ✓ Condensation
- Fusion
- Evaporation
- Sublimation

Q(v). The vapour pressure of water at 100 degrees C is:

- 260 mmHg
- ✓ 760 mmHg
- 1074 mmHg
- 1489 mmHg

Q(vi). The boiling point of glycerin at 50 mmHg is:

- 100 degrees C
- ✓ 210 degrees C
- 240 degrees C
- 360 degrees C

Q(vii). Food cooks more rapidly at:

- ✓ Karachi
- Quetta
- Murree
- Mount Everest

Q(viii). The SI unit of surface tension is:

- Dyne/cm
- Erg/cm^2
- Joule/litre
- ✓ Newton/metre

Q(ix). Viscosity of a liquid increases with a decrease in:

- Molecular mass
- Intermolecular forces

✓ Temperature

Atmospheric pressure

Q(x). The more volatile substance is:

Water

Methyl alcohol

Ethanol

✓ Diethyl ether

Exercise - Restricted Response Questions (RRQs)

Q(i). Evaporation is a cooling process and occurs at all temperatures. How?

Ans. Evaporation is a cooling process because the molecules that escape from the surface are the ones with the highest kinetic energy; their departure leaves the remaining liquid with a lower average kinetic energy, hence a lower temperature. It occurs at all temperatures because, at any temperature, some surface molecules have enough energy to overcome the intermolecular forces and escape, even though the liquid is not boiling — the fraction of such molecules is simply larger at higher temperatures.

Q(ii). Why do wet clothes hung outdoors dry rapidly on a hot summer day than on a cold winter day? Explain.

Ans. On a hot summer day the temperature is high, so more water molecules gain enough energy to evaporate, and the dry, moving air carries the vapour away, keeping the evaporation rate high. On a cold winter day the molecules have less kinetic energy, fewer escape, and the air is often still and humid, so evaporation is slow.

That is why wet clothes dry rapidly in hot summer weather.

Q(iii). Why does steam at 100 degrees C burn your skin much more severely than liquid water at 100 degrees C?

Ans. Steam at 100 degrees C burns more severely than liquid water at 100 degrees C because steam carries extra energy: the latent heat of vaporisation (about 2260 kJ/kg). When steam touches the skin it first condenses, releasing this latent heat, and then the condensed water at 100 degrees C gives up further heat as it cools. Liquid water at 100 degrees C only gives up heat by cooling, so steam delivers much more energy to the skin.

Q(iv). Why do droplets appear outside the glass full of ice-cold water in the summertime?

Ans. The cold glass cools the air next to it below its dew point, so the water vapour in the warm, humid summer air condenses on the outside of the glass. These tiny condensed droplets are what we see as droplets on the glass; the water comes from the air, not from inside the glass.

Q(v). How can a razor blade float on the surface of water, despite being denser than water?

Ans. A razor blade floats because of surface tension: the surface of water behaves like a stretched elastic skin due to the attraction between surface water molecules. If the blade is placed gently and horizontally, the surface tension supports its weight, even though steel is denser than water. The blade does not break the surface, so it rests on the 'skin' of the water; if it is pushed through, it sinks at

once.

Q(vi). Why does a small drop of water assume nearly a spherical shape on the surface of the waxy bonnet of the car?

Ans. A small drop of water on a waxy car bonnet becomes nearly spherical because water molecules attract each other (cohesion) much more strongly than they are attracted to the wax (adhesion). Surface tension then pulls the drop into the shape with the smallest surface area for its volume, which is a sphere. On a clean glass surface, adhesion is stronger, so the water spreads out instead.

Exercise - Extended Response Questions (ERQs)

Q(i). Explain the boiling point of the liquid. What are the factors affecting the boiling point of a liquid? How does the boiling point of a liquid depend on external pressure?

Ans. The boiling point of a liquid is the temperature at which its vapour pressure becomes equal to the external (atmospheric) pressure, so that bubbles can form throughout the liquid. Factors affecting it: (1) Intermolecular forces — stronger forces (hydrogen bonding) give higher boiling points. (2) Molecular mass and size — larger molecules have stronger London forces and higher boiling points. (3) External pressure — the boiling point depends directly on it: at higher external pressure, a higher temperature is needed for the vapour pressure to equal it, so the boiling point rises (food cooks faster in a pressure cooker, at higher pressure); at lower pressure (mountains), the boiling point falls and food cooks slowly. (4) Impurities — non-volatile solutes raise the boiling point.

Q(ii). Define crystalline and amorphous solids. How does a crystalline solid differ from an amorphous solid? Give examples of each.

Ans. Crystalline solids have a regular, ordered, three-dimensional arrangement of particles (a lattice), a sharp and definite melting point, and flat faces with definite angles; examples are NaCl, diamond, sugar and ice. Amorphous solids have no long-range order — their particles are arranged irregularly, like a frozen liquid; they soften gradually over a range of temperature and have no sharp melting point; examples are glass, rubber, plastic and pitch. Differences: crystalline solids are anisotropic (properties differ with direction) and have definite geometry, while amorphous solids are isotropic and have irregular shapes; crystalline solids break along definite cleavage planes, while amorphous solids break irregularly.

Q(iii). Explain the following with reasons: (a) Solids have a very low vapour pressure compared to liquids. (b) Amorphous solids are termed 'supercooled liquids'. (c) Crystalline solids have sharp and distinct melting points, while amorphous solids do not.

Ans. (a) Solids have a very low vapour pressure compared to liquids because their particles are held by strong forces in fixed positions and vibrate only slightly; very few particles gain enough energy to escape from the surface, whereas liquid particles move freely and escape much more easily. (b) Amorphous solids are termed 'supercooled liquids' because they have no ordered crystal structure — their arrangement resembles that of a liquid frozen in place — and they flow extremely slowly over very long times (for example, old window glass is thicker at the bottom). (c) Crystalline

solids have sharp and distinct melting points because all their bonds are identical and arranged regularly, so the whole lattice breaks down at one definite temperature; amorphous solids have bonds of varying strengths in an irregular arrangement, so they soften gradually over a temperature range instead of melting sharply.

Unit 5: Chemical Energetics

Exercise - Multiple Choice Questions (MCQs)

Q(i). The branch of chemistry that deals with heat changes during a chemical reaction is:

- Thermodynamics
- Electrochemistry
- ✓ Thermochemistry
- Biochemistry

Q(ii). A positive standard entropy change (delta S) indicates:

- ✓ An increase in the randomness or disorder of the system
- A decrease in the randomness or disorder of the system
- No change in the disorder of the system
- That the reaction must be exothermic

Q(iii). In an exothermic reaction, the sign of delta H is:

- Positive
- ✓ Negative
- Zero
- Variable

Q(iv). Under standard conditions, a reaction is spontaneous when:

- ✓ Gibbs free energy change (delta G) is negative
- Gibbs free energy change (delta G) is positive
- The enthalpy change (delta H) is positive and the entropy

change (delta S) is negative

Temperature is zero Kelvin

Q(v). The equation used to calculate the heat absorbed or released during a temperature change of a substance is:

✓ q = mc delta T

m = delta H

delta G = delta H - T delta S

delta H = n

Q(vi). A positive delta S value indicates that the solution:

Becomes more ordered

Absorbs heat

✓ Becomes more disordered

Releases heat

Q(vii). The measure of the amount of energy released when an electron is added to the lowest energy state of a gaseous atom to form an anion is:

Ionisation energy

✓ Electron affinity

delta G = (delta H - T delta S)

Bond dissociation energy

Q(viii). Entropy is associated with:

The amount of heat absorbed or released

✓ The disorder or randomness of a system

The pressure of a gas

The volume of a substance

Q(ix). The formula for calculating the standard enthalpy change of a reaction (delta H) is:

delta H(rxn) = delta S(rxn) - T delta G(rxn)

✓ delta H(rxn) = delta G(rxn) + T delta S(rxn)

delta H(rxn) = delta G(rxn) - T delta S(rxn)

delta H(rxn) = delta G(rxn) / T delta S(rxn)

Q(x). The condition that indicates a spontaneous reaction under standard conditions is:

delta G > 0

✓ delta G < 0

delta H < 0 and delta S > 0

delta H > 0 and delta S < 0

Exercise - Restricted Response Questions (RRQs)

Q(i). Define activation energy and its role in chemical reactions.

Ans. Activation energy is the minimum extra energy that reactant particles must gain before they can react — the energy needed to reach the activated complex (transition state) at the top of the energy barrier. Its role: it controls the rate of a reaction — a high activation energy means few particles have enough energy, so the reaction is slow; a low activation energy means the reaction is fast. Catalysts work by providing a path with a lower activation energy.

Q(ii). What is the heat of reaction, and what does it indicate?

Ans. The heat of reaction (enthalpy change, delta H) is the heat

absorbed or released when a reaction occurs at constant pressure, for the amounts shown in the balanced equation. Its sign indicates the direction of heat flow: a negative delta H means the reaction releases heat (exothermic — products have less enthalpy than reactants), while a positive delta H means the reaction absorbs heat (endothermic — products have more enthalpy than reactants).

Q(iii). What are the standard conditions for standard entropy change?

Ans. Standard conditions for standard entropy change are: all substances in their standard states — pure solids and liquids, gases at 1 atmosphere (1 bar) pressure, and solutions at 1 mol/dm^3 concentration — at a stated temperature, usually 298 K (25 degrees C). Standard entropy values (S) are measured relative to a perfect crystal at 0 K, where entropy is zero.

Q(iv). What is the standard enthalpy change of a reaction?

Ans. The standard enthalpy change of a reaction (delta H) is the heat change when the reaction occurs with all reactants and products in their standard states (1 atm, 298 K, 1 mol/dm^3 for solutions), for the molar amounts in the balanced equation. It is found by Hess's law: delta H = sum of standard enthalpies of formation of products minus sum of standard enthalpies of formation of reactants.

Q(v). Explain the role of Gibbs free energy in predicting reaction spontaneity.

Ans. Gibbs free energy (G) combines enthalpy and entropy: delta G

= delta H - T delta S. It predicts spontaneity at constant temperature and pressure: if delta G is negative, the reaction is spontaneous (feasible); if delta G is positive, it is non-spontaneous; if delta G is zero, the system is at equilibrium. So an exothermic reaction with increasing disorder is always spontaneous, while an endothermic reaction with decreasing disorder is never spontaneous; the other two combinations depend on temperature.

Exercise - Extended Response Questions (ERQs)

Q(i). Define Thermochemistry and explain the difference between exothermic and endothermic reactions. Provide examples of each.

Ans. Thermochemistry is the branch of chemistry that studies the heat changes accompanying chemical reactions. In an exothermic reaction, heat is released to the surroundings, delta H is negative, and the products have less enthalpy than the reactants — examples are combustion (burning methane), freezing and condensation. In an endothermic reaction, heat is absorbed from the surroundings, delta H is positive, and the products have more enthalpy than the reactants — examples are photosynthesis, melting of ice and evaporation of water. The difference is the direction of heat flow and the sign of delta H.

Q(ii). Explain the relationship between activation energy and the spontaneity of a chemical reaction. How does activation energy differ for exothermic and endothermic reactions? Draw energy diagrams for exothermic and endothermic reactions.

Ans. Activation energy does not decide spontaneity — it decides rate. A spontaneous reaction (negative delta G) may still be very

slow if its activation energy is high (for example, the combustion of paper in air is spontaneous but needs a spark). For an exothermic reaction, the activation energy of the forward reaction is smaller than that of the reverse reaction, because the products lie lower in energy; the energy diagram starts high (reactants), rises to a peak (activated complex) and falls to a lower level (products), with the drop equal to delta H. For an endothermic reaction, the forward activation energy is larger than the reverse activation energy; the diagram rises to a peak and ends higher than it started, with the rise equal to delta H. A catalyst lowers the activation energy for both directions equally, speeding the reaction without changing delta H or spontaneity.

Q(iii). Explain how to predict the spontaneity of a reaction based on the signs of H and S. Provide examples to support your explanation.
Ans. Spontaneity is predicted from delta G = delta H - T delta S. Four cases: (1) delta H negative, delta S positive: delta G always negative — spontaneous at all temperatures (for example, combustion of fuels). (2) delta H positive, delta S negative: delta G always positive — never spontaneous (for example, the decomposition of CO2 into C and O2 at ordinary temperatures). (3) delta H negative, delta S negative: spontaneous only at low temperatures, where the enthalpy term dominates (for example, the Haber process, N2 + 3H2 -> 2NH3, delta H negative, delta S negative, favoured at low temperature). (4) delta H positive, delta S positive: spontaneous only at high temperatures, where the T delta S term dominates (for example, melting of ice, or the decomposition of CaCO3 on heating).

Exercise - Numerical Problems

Q(i). The standard enthalpy of formation of MgO(s) is -601 kJ/mol. Enthalpy of sublimation of Mg(s) = +148 kJ/mol; First ionisation energy of Mg(g) = +738 kJ/mol; Second ionisation energy of Mg+(g) = +1450 kJ/mol; Bond dissociation enthalpy of O2(g) = +498 kJ/mol; First electron affinity of O(g) = -141 kJ/mol; Second electron affinity of O-(g) = +798 kJ/mol. Using Hess's Law and the Born-Haber cycle, calculate the lattice energy of MgO(s).

Step 1: Born-Haber cycle: delta Hf = sublimation + IE1 + IE2 + (1/2) bond dissociation of O2 + EA1 + EA2 + lattice energy.

Step 2: Substituting: -601 = 148 + 738 + 1450 + 249 - 141 + 798 + lattice energy.

Step 3: Sum of the known terms = 3242 kJ/mol, so lattice energy = -601 - 3242 = -3843 kJ/mol from the values as printed; the book gives the lattice energy as -3886 kJ/mol (a difference of 43 kJ/mol, from the reference values used in the book).

Answer: Lattice energy of MgO(s) = -3886 kJ/mol (book answer; see note in the manifest).

Q(ii). A chemical reaction involves 100 g of a substance with a specific heat capacity of 2.50 J/g/degree C. The temperature increases by 20 degrees C, and 0.25 mol of the substance is involved. (a) Calculate the heat (q) absorbed or released using q = mc delta T. (b) Calculate the enthalpy change per mole (delta H)

using delta H = -mc delta T / n. (c) State whether the reaction is endothermic or exothermic.

Step 1: (a) q = m x c x delta T = 100 g x 2.50 J/g/degree C x 20 degree C = 5000 J = 5.00 kJ.

Step 2: (b) delta H = -mc delta T / n = -5000 J / 0.25 mol = -20,000 J/mol = -20.0 kJ/mol.

Step 3: (c) The negative sign of delta H shows heat is released, so the reaction is exothermic.

Answer: (a) q = 5.00 kJ. (b) delta H = -20.0 kJ/mol. (c) Exothermic.

Q(iii). The average bond energies for H-H is 436 kJ/mol, Cl-Cl is 243 kJ/mol, and H-Cl is 431 kJ/mol. Calculate the enthalpy change (delta H) for the reaction: H2(g) + Cl2(g) -> 2HCl(g).

Step 1: Energy needed to break bonds (reactants) = H-H + Cl-Cl = 436 + 243 = 679 kJ/mol.

Step 2: Energy released when bonds form (products) = 2 x H-Cl = 2 x 431 = 862 kJ/mol.

Step 3: delta H = bonds broken - bonds formed = 679 - 862 = -183 kJ/mol.

Answer: delta H = -183 kJ/mol.

Q(iv). The enthalpy change for the complete combustion of ethane

(C2H6) is -1560 kJ/mol. The standard enthalpies of formation for CO2(g) and H2O(l) are -393.5 kJ/mol and -285.8 kJ/mol, respectively. The standard enthalpy of formation for O2(g) is zero. Using Hess's Law, calculate the standard enthalpy of formation (delta Hf) of ethane. The balanced combustion equation is: C2H6(g) + 7/2 O2(g) -> 2CO2(g) + 3H2O(l).

Step 1: Hess's law: delta H combustion = sum of delta Hf of products - sum of delta Hf of reactants.

Step 2: -1560 = [2 x (-393.5) + 3 x (-285.8)] - [delta Hf(C2H6) + 0].

Step 3: -1560 = (-787 - 857.4) - delta Hf(C2H6) = -1644.4 - delta Hf(C2H6).

Step 4: delta Hf(C2H6) = -1644.4 + 1560 = -84.4 kJ/mol.

Answer: delta Hf of ethane = -84.4 kJ/mol.

Q(v). For the reaction: N2(g) + 3H2(g) -> 2NH3(g), delta H = -92.2 kJ/mol, delta S = -198 J/mol/K, and T = 298 K. Calculate delta G and predict whether the reaction is spontaneous under standard conditions.

Step 1: Convert delta S to kJ: -198 J/mol/K = -0.198 kJ/mol/K.

Step 2: delta G = delta H - T delta S = -92.2 - (298 x -0.198) = -92.2 + 59.0 = -33.2 kJ/mol.

Step 3: Since delta G is negative, the reaction is spontaneous under

standard conditions.

Answer: delta G = -33.2 kJ/mol; the reaction is spontaneous.

Self-Assessment (model answers)

Calculate the entropy change for each of these processes: (a) 2CO2(g) -> 2CO(g) + O2(g). (b) NaCl(s) -> NaCl(aq). The S values for CO2, CO, O2, NaCl(s), and NaCl(aq) are 213.7, 197.7, 205.1, 72.1 and 115.5 J/mol/K respectively.

Step 1: (a) delta S = [2 x S(CO) + S(O2)] - [2 x S(CO2)] = [2 x 197.7 + 205.1] - [2 x 213.7] = 600.5 - 427.4 = 173.1 J/mol/K.

Step 2: (b) delta S = S(NaCl, aq) - S(NaCl, s) = 115.5 - 72.1 = 43.4 J/mol/K.

Answer: (a) delta S = 173.1 J/mol/K. (b) delta S = 43.4 J/mol/K.

Unit 6: Reaction Kinetics

Exercise - Multiple Choice Questions (MCQs)

Q(i). Chemical kinetics is the branch of chemistry concerned with:

- Measuring equilibrium constants
- Determining the heat change in reactions
- ✓ Studying the rate and mechanism of chemical reactions
- Analysing the composition of chemical compounds

Q(ii). The energy of the activated complex is:

- Lower than the energy of reactants and higher than the energy of products
- Lower than the energy of products and higher than the energy of reactants
- Lower than the energy of both the reactants and products
- ✓ Higher than the energy of both the reactants and products

Q(iii). A substance that alters the rate of a chemical reaction without being consumed in the reaction is called:

- ✓ Catalyst
- Reactant
- Product
- Activated complex

Q(iv). A catalyst that slows down a chemical reaction is called:

- Positive catalyst

✓ Negative catalyst
Poison
Promoter

Q(v). The rate at which the reaction increases four times by doubling the concentration(s) of reactant(s) is:

First order
✓ Second order
Third order
Zero order

Q(vi). The decomposition of N_2O_5 is an example of:

✓ First order
Second order
Third order
Zero order

Q(vii). The half-life of a first-order reaction is:

Directly proportional to initial concentration
✓ Independent of initial concentration
Inversely proportional to the square of concentration
Dependent on reaction mechanism only

Q(viii). The factor that affects the rate of a chemical reaction is:

Nature of the product
Shape of the container
✓ Catalyst
Volume of the container

Q(ix). The unit of the rate constant for a third-order reaction is:

- s^-1
- mol^-1 dm^3 s^-1
- ✓ mol^-2 dm^6 s^-1
- mol dm^-3 s^-1

Q(x). The following reaction is NOT an example of heterogeneous catalysis:

- 2SO2(g) + O2(g) --Pt--> 2SO3(g)
- CH2=CH2(g) + H2(g) --Ni--> CH3-CH3(g)
- CO(g) + 2H2(g) --Cr2O3+ZnO--> CH3OH(g)
- ✓ 2SO2(g) + O2(g) --NO--> 2SO3(g)

Exercise - Restricted Response Questions (RRQs)

Q(i). Define the rate-determining step and elementary step.

Ans. An elementary step is a single step in a reaction mechanism — one collision event that directly converts reactants into products or intermediates. The rate-determining step is the slowest elementary step in a multi-step mechanism; because the overall reaction cannot go faster than its slowest step, this step controls the rate of the whole reaction and determines the rate law.

Q(ii). What is the effect of a catalyst on activation energy?

Ans. A catalyst lowers the activation energy of a reaction by providing an alternative reaction pathway with a lower energy barrier. It lowers the barrier for the forward and reverse reactions equally, so it speeds up the reaction in both directions but does not

change the enthalpy change, the equilibrium constant or the yield; it only helps equilibrium to be reached faster.

Q(iii). How can the rate of a reaction increase without changing the temperature?

Ans. The rate can be increased without changing the temperature by: increasing the concentration of reactants (more frequent collisions); increasing the surface area of a solid reactant (powdering it); using a catalyst (lowering activation energy); or, for gases, increasing the pressure (which increases concentration). Stirring and the nature of the reactants also affect the rate.

Q(iv). Why is the order of a reaction not necessarily equal to the sum of the coefficients in the balanced chemical equation?

Ans. The order of a reaction is determined experimentally from the rate law and depends on the mechanism, especially on the rate-determining step; it is the sum of the powers of concentration terms in the rate law. The balanced equation only shows the overall stoichiometry, which may involve several elementary steps; only the species in the slow step (and those in fast steps before it) appear in the rate law. So the order need not equal the sum of the stoichiometric coefficients — for example, the reaction H2 + Br2 -> 2HBr has a complex rate law, not simple second order.

Q(v). Why is radioactive decay considered a first-order reaction?

Ans. Radioactive decay is a first-order reaction because the rate of decay depends only on the number of undecayed nuclei present at that instant: rate = kN, where N is the number of nuclei. Each

nucleus has the same probability of decaying in a given time, independent of the others, so doubling the amount doubles the rate — the defining feature of first order. Its half-life is therefore constant and independent of the initial amount.

Exercise - Extended Response Questions (ERQs)

Q(i). What is chemical kinetics? What information does it provide about chemical reactions?

Ans. Chemical kinetics is the study of the rates of chemical reactions, the factors that affect them, and the step-by-step mechanisms by which reactions occur. It provides: the rate law (how rate depends on concentrations); the order of reaction and the rate constant; the half-life of reactants; the activation energy and the effect of temperature (Arrhenius equation); and the mechanism — the sequence of elementary steps, including which step is rate-determining. This information lets chemists control reaction speeds in industry and understand how reactions actually happen at the molecular level.

Q(ii). What is a catalyst? What are the characteristics of a catalyst?

Ans. A catalyst is a substance that increases the rate of a chemical reaction without itself being consumed, by providing an alternative pathway of lower activation energy. Characteristics: it is not used up and can be recovered unchanged at the end; a small amount can catalyse a large amount of reactant; it does not change the equilibrium position or the enthalpy change of the reaction; it is specific — a catalyst for one reaction may not work for another; and it can be poisoned (deactivated) by impurities. A positive

catalyst speeds a reaction up; a negative catalyst (inhibitor) slows it down.

Q(iii). Provide an example for each of first-, second-, third-, and zero-order chemical reactions.
Ans. First-order: the decomposition of N2O5 (2N2O5 -> 4NO2 + O2) and radioactive decay — rate depends on one concentration term to the first power. Second-order: the reaction 2NO2 -> 2NO + O2, or H2 + I2 -> 2HI — rate depends on the square of one concentration or on two first-power concentrations. Third-order: the reaction 2NO + O2 -> 2NO2 — rate = k[NO]^2[O2]. Zero-order: the decomposition of ammonia on a hot platinum surface, or photochemical reactions such as H2 + Cl2 in sunlight — the rate is independent of concentration because the surface (or light) is saturated.

Exercise - Numerical Problem

Q(i). Consider the formation of phosgene gas from CO and Cl2: CO(g) + Cl2(g) -> COCl2(g). The initial rates obtained from three experiments at a given temperature are: Expt 1: [CO] = 0.015, [Cl2] = 0.0068, rate = 1.7 x 10^-10 mol/dm^3/s. Expt 2: [CO] = 0.030, [Cl2] = 0.0068, rate = 3.40 x 10^-10 mol/dm^3/s. Expt 3: [CO] = 0.015, [Cl2] = 0.0204, rate = 5.10 x 10^-10 mol/dm^3/s. Using the data, determine (a) the reaction orders for both reactants, the overall order of reaction, and the rate law; and (b) the value of the rate constant at the specified temperature.
Step 1: Compare experiments 1 and 2: [Cl2] is constant, [CO] doubles (0.015 to 0.030) and the rate doubles (1.7 to 3.40 x

10^-10). So the reaction is first order with respect to CO.

Step 2: Compare experiments 1 and 3: [CO] is constant, [Cl2] triples (0.0068 to 0.0204) and the rate triples (1.7 to 5.10 x 10^-10). So the reaction is first order with respect to Cl2.

Step 3: Overall order = 1 + 1 = 2 (second order). Rate law: rate = k[CO][Cl2].

Step 4: From experiment 1: k = rate / ([CO][Cl2]) = 1.7 x 10^-10 / (0.015 x 0.0068) = 1.7 x 10^-10 / 1.02 x 10^-4 = 1.67 x 10^-6 dm^3 mol^-1 s^-1.

Answer: First order in CO, first order in Cl2; overall second order; rate = k[CO][Cl2]; k = 1.67 x 10^-6 dm^3 mol^-1 s^-1.

Unit 7: Acid, Base and Chemical Equilibrium

Exercise - Multiple Choice Questions (MCQs)

Q(i). In the Contact Process, the primary product formed is:

- Nitrogen
- Ammonia
- ✓ Sulphur trioxide
- Carbon dioxide

Q(ii). The unit of Kc for the reaction: 2SO2(g) + O2(g) <=> 2SO3(g) is:

- mol dm^-3
- mol^-1 dm^-3
- ✓ mol^-1 dm^3
- mol^2 dm^-6

Q(iii). In a reaction mixture, the ratio of the concentrations of products to reactants at equilibrium is known as:

- Reaction quotient
- ✓ Equilibrium constant
- Rate constant
- Activation energy

Q(iv). Exothermic reactions are favoured in the forward direction by:

- ✓ Decrease in temperature
- Increase in temperature

Adding a catalyst

Removal of reactant

Q(v). The main purpose of the Haber process is to:

Produce sulfuric acid

✓ Synthesise ammonia

Create nitrogen gas

Manufacture hydrogen gas

Q(vi). The indicator that best suits a titration between a weak acid and a strong base is:

Methyl orange

Methyl red

✓ Phenolphthalein

Bromothymol blue

Q(vii). The value of the ionic product of water at 25 degrees C is:

✓ 1.0 x 10^-14 mol^2 dm^-6

1.0 x 10^14 mol^2 dm^-6

1.5 x 10^14 mol^2 dm^-6

1.5 x 10^-14 mol^2 dm^-6

Q(viii). When an acid is added to water, then:

✓ [H+] > [OH-]

[H+] < [OH-]

[H+] = [OH-]

[H+] approx [OH-]

Q(ix). The pH of a buffer solution prepared with 1M acetic acid (CH3COOH) and 1M acetate (CH3COO-) is:

- pH = 2.37
- ✓ pH = 4.74
- pH = 7.00
- pH = 10.26

Q(x). The suitable indicator for a titration involving a weak acid and a strong base is:

- ✓ Phenolphthalein
- Methyl orange
- Bromothymol blue
- Litmus paper

Exercise - Restricted Response Questions (RRQs)

Q(i). What does Le Chatelier's Principle state?

Ans. Le Chatelier's Principle states: if a change of condition (concentration, pressure or temperature) is applied to a system at equilibrium, the system shifts in the direction that tends to undo (oppose) the change, establishing a new equilibrium.

Q(ii). What happens to the equilibrium when the concentration of reactants is increased?

Ans. When the concentration of reactants is increased, the equilibrium shifts to the right (towards the products), because the system uses up the added reactant to oppose the change. More product forms until a new equilibrium is reached. (Adding product shifts the equilibrium to the left.)

Q(iii). What effect does temperature have on an exothermic reaction?

Ans. For an exothermic reaction, increasing the temperature shifts the equilibrium to the left (towards the reactants), because the system absorbs the added heat by favouring the endothermic reverse reaction. Decreasing the temperature shifts it to the right, favouring the exothermic forward reaction. Temperature is the only factor that changes the value of the equilibrium constant.

Q(iv). What is the effect of removing a product from an equilibrium mixture?

Ans. Removing a product from an equilibrium mixture shifts the equilibrium to the right (towards the products), because the system replaces the removed product by converting more reactants. This is used in industry — for example, removing ammonia from the Haber process mixture increases the yield.

Q(v). What is the significance of pKa in acid-base chemistry?

Ans. pKa = -log Ka, where Ka is the acid dissociation constant. It measures acid strength on a convenient scale: a smaller pKa means a larger Ka and a stronger acid. pKa is used to compare acid strengths, to predict the pH of buffer solutions (pH = pKa when acid and conjugate base concentrations are equal), and to choose a suitable indicator for a titration.

Q(vi). Name two indicators commonly used in acid-base titrations.

Ans. Two indicators commonly used in acid-base titrations are

methyl orange (red in acid, yellow in alkali, range about pH 3.1-4.4, suitable for strong acid-weak base titrations) and phenolphthalein (colourless in acid, pink in alkali, range about pH 8.3-10.0, suitable for weak acid-strong base titrations).

Q(vii). The aqueous solution of NH4Cl is acidic and that of Na2CO3 is basic, why?

Ans. NH4Cl is a salt of a weak base (NH4OH) and a strong acid (HCl): its NH4+ ion hydrolyses in water (NH4+ + H2O <=> NH4OH + H+), producing H+ ions, so the solution is acidic. Na2CO3 is a salt of a strong base (NaOH) and a weak acid (H2CO3): its CO3^2- ion hydrolyses (CO3^2- + H2O <=> HCO3- + OH-), producing OH- ions, so the solution is basic.

Q(viii). AlCl3 is said to be a Lewis acid but not a Bronsted-Lowry acid, why?

Ans. AlCl3 is a Lewis acid because the aluminium atom has an incomplete octet (only six electrons) and can accept a lone pair of electrons from a donor (for example, from Cl- to form AlCl4-). It is not a Bronsted-Lowry acid because it has no hydrogen (proton) to donate; Bronsted-Lowry acids must be proton donors, while Lewis acids are electron-pair acceptors, a wider definition that includes AlCl3.

Exercise - Extended Response Questions (ERQs)

Q(i). What microscopic events occur in a chemical system at equilibrium? Provide specific examples.

Ans. At equilibrium a system only looks still: microscopically, the

forward and reverse reactions continue at equal rates, so the concentrations of reactants and products stay constant. Examples: in the reaction H2 + I2 <=> 2HI, HI molecules keep forming and breaking apart at the same rate; in a saturated solution, solid keeps dissolving at the same rate as dissolved solute crystallises out; and in a closed bottle of water, evaporation and condensation proceed at equal rates, keeping the vapour pressure constant.

Q(ii). What is Le Chatelier's Principle, and how does it apply to concentration, pressure, and temperature changes?

Ans. Le Chatelier's Principle: a system at equilibrium opposes any change imposed on it. Concentration: adding a reactant (or removing a product) shifts the equilibrium towards the products; removing a reactant (or adding a product) shifts it towards the reactants. Pressure: increasing the pressure shifts the equilibrium towards the side with fewer gas molecules; decreasing it shifts it towards the side with more gas molecules (no effect if both sides have equal gas molecules). Temperature: increasing the temperature favours the endothermic direction; decreasing it favours the exothermic direction. A catalyst speeds both directions equally and does not shift the equilibrium position.

Q(iii). How does pressure affect gaseous equilibria?

Ans. Pressure affects only gaseous equilibria, and only when the two sides have different numbers of gas molecules. Increasing the pressure shifts the equilibrium towards the side with fewer moles of gas, because that side occupies less volume and so opposes the pressure increase. Decreasing the pressure shifts it towards the

side with more moles of gas. Example: in N2 + 3H2 <=> 2NH3 (4 moles of gas on the left, 2 on the right), high pressure favours ammonia formation. In H2 + I2 <=> 2HI (2 moles on each side), pressure has no effect on the equilibrium position.

Q(iv). Describe the process of salt hydrolysis and provide an example of a salt that undergoes hydrolysis.

Ans. Salt hydrolysis is the reaction of the ions of a salt with water, producing an acidic or basic solution. It occurs when the salt comes from a weak acid or a weak base (or both): the ion of the weak partner reacts with water to give the undissociated weak acid or base plus H+ or OH-. Example: ammonium chloride, NH4Cl, hydrolyses as NH4+ + H2O <=> NH4OH + H+, giving an acidic solution (pH < 7). Salts of strong acids and strong bases (for example NaCl) do not hydrolyse, and their solutions are neutral.

Q(v). How does pressure affect gaseous equilibria?

Ans. (This repeats ERQ (iii) in the book.) Pressure affects only gaseous equilibria with unequal gas moles on the two sides: increasing pressure shifts the equilibrium towards the side with fewer gas molecules, and decreasing pressure shifts it towards the side with more gas molecules. If both sides have the same number of gas molecules, pressure has no effect. Example: in the Haber process, N2 + 3H2 <=> 2NH3, high pressure favours the forward reaction (4 gas moles -> 2 gas moles), which is why the industrial process runs at about 200 atmospheres.

Q(vi). Explain the common ion effect and how it influences the

solubility of salts. Provide an example to support your explanation.
Ans. The common ion effect is the decrease in the solubility (or ionisation) of a weak electrolyte when a strong electrolyte containing a common ion is added. The added common ion shifts the equilibrium towards the undissociated form. Example: acetic acid ionises as CH3COOH <=> CH3COO- + H+; adding sodium acetate (which supplies CH3COO-) pushes the equilibrium to the left, reducing the ionisation of acetic acid. Similarly, the solubility of AgCl in water decreases when NaCl is added, because the extra Cl- ions shift AgCl(s) <=> Ag+ + Cl- to the left, precipitating more AgCl. The effect is used in salt purification and in buffer solutions.

Exercise - Numerical Problems
Q(i). The equilibrium concentrations of hydrogen, iodine, and hydrogen iodide are 0.530 M, 0.033 M, and 0.934 M respectively at 450 degrees C. What is the value of Kc for the following reaction? H2(g) + I2(g) <=> 2HI(g).
Step 1: Kc = [HI]^2 / ([H2][I2]).

Step 2: Substituting: Kc = (0.934)^2 / (0.530 x 0.033) = 0.8724 / 0.01749.

Step 3: Kc = 49.86.

Answer: Kc = 49.86.

Q(ii). Calculate the concentration of PCl5 at 250 degrees C. The equilibrium concentration for both PCl3 and Cl2 gases are 0.005 M.

The value of Kc for the reaction is 1.2 x 10^-2. PCl5(g) <=> PCl3(g) + Cl2(g).

Step 1: Kc = [PCl3][Cl2] / [PCl5].

Step 2: Rearranging: [PCl5] = [PCl3][Cl2] / Kc = (0.005 x 0.005) / (1.2 x 10^-2).

Step 3: [PCl5] = 2.5 x 10^-5 / 1.2 x 10^-2 = 2.08 x 10^-3 M.

Answer: [PCl5] = 2.08 x 10^-3 M.

Q(iii). Calculate the pH of each of the following solutions and classify each solution as acidic, basic or neutral: (a) pOH = 0.00 (b) pOH = 1.62 (c) pOH = 3.33 (d) pOH = 7.00 (e) pOH = 13.92.

Step 1: Use pH + pOH = 14, so pH = 14 - pOH.

Step 2: (a) pH = 14 - 0.00 = 14.00 (basic). (b) pH = 14 - 1.62 = 12.38 (basic). (c) pH = 14 - 3.33 = 10.67 (basic). (d) pH = 14 - 7.00 = 7.00 (neutral). (e) pH = 14 - 13.92 = 0.08 (acidic).

Answer: (a) pH 14.00, basic. (b) pH 12.38, basic. (c) pH 10.67, basic. (d) pH 7.00, neutral. (e) pH 0.08, acidic.

Q(iv). Calculate the pOH of each of the following solutions and classify each solution as acidic, basic or neutral: (a) pH = 14.02 (b) pH = 12.14 (c) pH = 7.61 (d) pH = 7.00 (e) pH = 11.44.

Step 1: Use pOH = 14 - pH.

Step 2: (a) pOH = 14 - 14.02 = -0.02. (b) pOH = 14 - 12.14 = 1.86. (c) pOH = 14 - 7.61 = 6.39. (d) pOH = 14 - 7.00 = 7.00. (e) pOH = 14 - 11.44 = 2.56.

Step 3: Classification by pH: (a) 14.02 basic, (b) 12.14 basic, (c) 7.61 basic, (d) 7.00 neutral, (e) 11.44 basic.

Answer: (a) pOH -0.02, basic. (b) pOH 1.86, basic. (c) pOH 6.39, basic. (d) pOH 7.00, neutral. (e) pOH 2.56, basic.

Q(v). What is pOH of the solution having hydrogen ion concentration 1.58 x 10^-2 M? Is the solution acidic or basic?

Step 1: pH = -log[H+] = -log(1.58 x 10^-2) = 1.80.

Step 2: pOH = 14 - pH = 14 - 1.80 = 12.20.

Step 3: Since pH < 7, the solution is acidic.

Answer: pOH = 12.20 (Acidic).

Q(vi). What is the pH of 2.0 x 10^-4 M NaOH solution?

Step 1: NaOH is a strong base, so [OH-] = 2.0 x 10^-4 M.

Step 2: pOH = -log(2.0 x 10^-4) = 3.70.

Step 3: pH = 14 - 3.70 = 10.30 (Basic).

Answer: pH = 10.30 (Basic).

Q(vii). Calculate the pH of 0.01 M acetic acid whose Ka value is 1.7 x 10^-5.

Step 1: For a weak acid, [H+] = sqrt(Ka x C) = sqrt(1.7 x 10^-5 x 0.01) = sqrt(1.7 x 10^-7) = 4.12 x 10^-4 M.

Step 2: pH = -log(4.12 x 10^-4) = 3.38 (Acidic).

Answer: pH = 3.38 (Acidic).

Q(viii). Formic acid has 2.45 x 10^-6 M hydrogen ion concentration and 2.45 x 10^-6 M formate ion concentration. The concentration of unionized formic acid is 0.025 M at equilibrium. Find out the value of Ka for the acid at room temperature.

Step 1: Ka = [H+][HCOO-] / [HCOOH].

Step 2: Ka = (2.45 x 10^-6 x 2.45 x 10^-6) / 0.025 = 6.00 x 10^-12 / 0.025.

Step 3: Ka = 2.4 x 10^-10.

Answer: Ka = 2.4 x 10^-10 (at room temperature).

Unit 8: Periodic Table and Periodicity

Exercise - Multiple Choice Questions (MCQs)

Q(i). Elements in the same group of the periodic table typically have similar:

- Atomic mass
- Number of neutrons
- ✓ Number of valence electrons
- Number of electron shells

Q(ii). The block of the periodic table that contains the transition metals is:

- s-block
- p-block
- ✓ d-block
- f-block

Q(iii). The element that is a metalloid is:

- Sodium (Na)
- Chlorine (Cl)
- ✓ Silicon (Si)
- Magnesium (Mg)

Q(iv). The description of electron affinity is:

- Energy required to remove an electron
- ✓ Energy released when an electron is added

Ability to attract electrons in a bond

Size of an atom

Q(v). The lanthanides and actinides are located in:

s-block

p-block

d-block

✓ f-block

Q(vi). The general electronic configuration of s-block elements is:

ns^2 np^1-6

(n-1)d^1-10 ns^1-2

✓ ns^1-2

(n-2)f^1-14 (n-1)d^0-1 ns^1-2

Q(vii). Aluminium hydroxide (Al(OH)3) is best described as:

A strong base

A strong acid

✓ Amphoteric

Neutral

Q(viii). The type of bonding mainly present in magnesium oxide (MgO) is:

Covalent

✓ Ionic

Metallic

Hydrogen

Q(ix). The element most likely to form a covalent bond is:

- Potassium (K)
- Calcium (Ca)
- ✓ Sulphur (S)
- Magnesium (Mg)

Q(x). A characteristic property of alkali metals is:

- High electronegativity
- ✓ Low ionisation energy
- High melting point
- Poor electrical conductivity

Exercise - Restricted Response Questions (RRQs)

Q(i). What causes trends in metallic and non-metallic behaviour?

Ans. The trends in metallic and non-metallic behaviour are caused by the periodic trends in atomic size, ionisation energy, electron affinity and electronegativity. Across a period, increasing nuclear charge pulls electrons closer, raising ionisation energy and electronegativity, so elements hold their electrons tightly and tend to gain electrons — they become non-metallic. Down a group, increasing size and shielding lower ionisation energy, so elements lose electrons easily and behave as metals.

Q(ii). What subshells are being filled in Group IIA elements?

Ans. In Group IIA (alkaline earth metals), the s subshell of the outermost shell is being filled: their general configuration is ns^2. For example, beryllium is 1s^2 2s^2, magnesium is [Ne] 3s^2 and calcium is [Ar] 4s^2. That is why they are s-block elements with

two valence electrons and a valency of 2.

Q(iii). What type of solution (acidic/basic) does AlCl3 form in water?
Ans. AlCl3 forms an acidic solution in water. The small, highly charged Al^3+ ion hydrolyses: [Al(H2O)6]^3+ loses a proton to water, giving [Al(H2O)5(OH)]^2+ + H3O+, so H+ ions are produced and the pH falls below 7.

Q(iv). What type of bonding is present in NaCl?
Ans. The bonding in NaCl is ionic: sodium, a metal with low ionisation energy, loses its single valence electron to form Na+, while chlorine, a non-metal with high electron affinity, gains it to form Cl-. The oppositely charged ions are held together by strong electrostatic attraction in a crystal lattice.

Q(v). Elements in Group IA are known to be highly reactive. What is the reason for this?
Ans. Group IA elements (alkali metals) are highly reactive because they have a single valence electron in a large s orbital, far from the nucleus and shielded by inner shells, so they have the lowest ionisation energies in their periods. They lose this electron very easily to form stable +1 ions with a noble-gas configuration, reacting vigorously with water, oxygen and halogens.

Exercise - Extended Response Questions (ERQs)
Q(i). Describe the key features of the arrangement of elements in the modern periodic table, including the importance of periods and groups.

Ans. In the modern periodic table, elements are arranged in order of increasing atomic number. The horizontal rows are periods (7 in total): the period number equals the number of electron shells (principal energy level) being filled. The vertical columns are groups (18 in total): elements in a group have the same number of valence electrons and therefore similar chemical properties. Importance: the group predicts valency and formula (for example, Group IA elements form +1 ions); the period predicts the number of shells; and the table is divided into s, p, d and f blocks according to which subshell is being filled, which explains why transition metals, lanthanides and actinides sit where they do.

Q(ii). Discuss how the position of an element in the periodic table can be used to determine its principal energy level and the number of valence electrons.

Ans. The period number of an element equals its highest principal energy level (number of shells): for example, sodium (period 3) has electrons in three shells. The group number (for main-group elements) gives the number of valence electrons: Group IA elements have 1, Group IIA have 2, Group VIIA have 7. For example, chlorine is in period 3 and group VIIA, so it has three shells and seven valence electrons, with configuration 1s^2 2s^2 2p^6 3s^2 3p^5. For transition metals, the group number relates to the total of ns and (n-1)d electrons.

Q(iii). Explain the trends in atomic radius, ionic radius, ionisation energy, electron affinity, and electronegativity as you move across a period and down a group in the periodic table.

Ans. Across a period (left to right): atomic and ionic radii decrease, because the nuclear charge increases while electrons enter the same shell, pulling the cloud inward; ionisation energy increases, because smaller atoms hold their electrons more tightly; electron affinity generally increases (becomes more negative) and electronegativity increases, for the same reason. Down a group: atomic and ionic radii increase, because new shells are added; ionisation energy decreases, because the outer electron is farther from the nucleus and more shielded; electron affinity decreases (with exceptions, such as fluorine being less than chlorine because of electron repulsion in its small 2p orbitals); and electronegativity decreases.

Q(iv). Explain the acidic and basic characters of sodium, magnesium, and aluminium oxides and hydroxides.

Ans. Sodium oxide and hydroxide are strongly basic: Na2O reacts with water to give NaOH, which ionises completely, and NaOH is a strong alkali. Magnesium oxide is also basic but less soluble: MgO with water gives Mg(OH)2, a weak base, because Mg^2+ holds its electrons more tightly and the hydroxide is only slightly soluble. Aluminium oxide and hydroxide are amphoteric: Al2O3 reacts with acids to give aluminium salts and water, and with bases to give aluminates; Al(OH)3 likewise dissolves in both acids and alkalis. The trend shows basic character decreasing across the period as the metal becomes less electropositive and the bond becomes more covalent.

Unit 9: p-Block Elements: Nitrogen, Sulphur, and Halogens

Exercise - Multiple Choice Questions (MCQs)

Q(i). In the Bronsted-Lowry theory, ammonia acts as a/an:

- Acid
- ✓ Base
- Salt
- Catalyst

Q(ii). The process of converting ammonia to nitrate under aerobic conditions is known as:

- Denitrification
- Nitrogen fixation
- ✓ Nitrification
- Ammonification

Q(iii). The acid known as the 'king of chemicals' is:

- Hydrochloric acid
- ✓ Sulphuric acid
- Nitric acid
- Acetic acid

Q(iv). Sulphuric acid is mainly prepared by which process?

- Haber process
- ✓ Contact process
- Solvay process

Ostwald process

Q(v). The catalyst used in selective catalytic reduction (SCR) for NOx removal is:

Rhodium

Platinum

✓ Vanadia-titania

Ammonia

Q(vi). The most reactive nonmetal halogen is:

Chlorine

✓ Fluorine

Bromine

Iodine

Q(vii). The electronic configuration of fluorine is:

✓ 1s^2 2s^2 2p^5

1s^2 2s^2 2p^6 3s^2 3p^5

1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 4p^5

1s^2 2s^2

Q(viii). The halogen that exists as a lustrous violet-black solid at room temperature is:

Chlorine

Bromine

Fluorine

✓ Iodine

Q(ix). The colour of the precipitate formed when chloride ions react with aqueous silver ions is:

✓ White

Cream-coloured

Yellow

Reddish brown

Q(x). The halogen that is a liquid under normal conditions is:

Fluorine

Chlorine

✓ Bromine

Iodine

Exercise - Restricted Response Questions (RRQs)

Q(i). Why does nitrogen, despite its abundance, not readily react with other substances at room temperature?

Ans. Nitrogen is unreactive at room temperature because the two nitrogen atoms in N2 are held by a very strong triple bond (one sigma and two pi bonds, bond energy about 945 kJ/mol). Breaking this bond needs a great deal of energy, and the molecule is non-polar, so it does not attract other molecules. Only extreme conditions (high temperature and pressure, as in the Haber process, or lightning) or catalysts can break the bond, which is why nitrogen, despite making up 78% of the air, does not readily react.

Q(ii). Differentiate between nitrification and denitrification in the nitrogen cycle.

Ans. Nitrification is the aerobic (oxygen-requiring) conversion of

ammonia to nitrite and then to nitrate by soil bacteria (Nitrosomonas converts NH3 to NO2-, Nitrobacter converts NO2- to NO3-); it makes nitrogen available to plants as nitrate. Denitrification is the opposite, anaerobic process: denitrifying bacteria (for example Pseudomonas) reduce nitrate back to nitrogen gas (N2), which returns to the atmosphere. Nitrification adds usable nitrogen to the soil; denitrification removes it.

Q(iii). What are the factors contributing to the low reactivity of sulphur?

Ans. The low reactivity of sulphur is due to: strong S-S single bonds that stabilise its ring (S8) and chain structures; low electron density and non-polar character, so it does not attract reactants strongly; multiple oxidation states, which let it form stable compounds rather than react further; and the stability of the compounds it forms, such as sulphides and sulphates. Like nitrogen, it usually needs heat or a catalyst to react.

Q(iv). What are the freezing point and boiling point of sulphuric acid?

Ans. Concentrated sulphuric acid freezes (melts) at about 10 degrees C (10.4 degrees C) and boils at about 337 degrees C. The high boiling point is due to strong hydrogen bonding between H2SO4 molecules; on boiling it partly decomposes into SO3 and water, so it does not have a sharp boiling point like a simple liquid.

Q(v). Why does dilute H2SO4 react with zinc but not with copper?

Ans. Dilute H2SO4 reacts with zinc because zinc is above hydrogen

in the reactivity series: it is more reactive than hydrogen, so it displaces hydrogen from the acid (Zn + H2SO4 -> ZnSO4 + H2). Copper is below hydrogen in the reactivity series: it is less reactive than hydrogen and cannot displace it from a dilute acid, so no reaction occurs with dilute H2SO4. (Concentrated H2SO4 does react with copper, as an oxidising acid, giving SO2.)

Q(vi). What black substance is formed when sugar is treated with conc. H2SO4?

Ans. The black substance is carbon. Concentrated sulphuric acid is a powerful dehydrating agent: it removes the elements of water (H and O) from sugar (sucrose, C12H22O11), leaving behind black carbon: C12H22O11 --conc. H2SO4--> 12C + 11H2O. The mixture also swells and steams as the reaction releases heat, which is why this is called the 'charring of sugar'.

Q(vii). What is the relationship between atomic size and volatility in halogens?

Ans. As the atomic size of halogens increases down the group (F < Cl < Br < I), the intermolecular (London) forces between molecules become stronger, so the boiling points rise and volatility falls: fluorine and chlorine are gases, bromine is a liquid, and iodine is a solid at room temperature. So larger halogen atoms mean lower volatility — the halogens become less volatile down the group as their size and molecular mass increase.

Q(viii). What is the maximum oxidation state exhibited by fluorine, and why is it unique in this regard?

Ans. The maximum oxidation state exhibited by fluorine is 0 (in F2) — or, in compounds, -1 only. Fluorine is unique because it is the most electronegative element: it can never show a positive oxidation state, since no element can take electrons away from it. Other halogens (Cl, Br, I) have vacant d orbitals and lower electronegativity, so they can expand their octet and show positive oxidation states up to +7 (for example in HClO4, BrO4- and IO4-), but fluorine, with only 2s and 2p orbitals and no d orbitals, cannot.

Q(ix). What is a disproportionation reaction? Provide an example involving chlorine.

Ans. A disproportionation reaction is one in which the same element is both oxidised and reduced at the same time. Example with chlorine: chlorine reacts with cold dilute sodium hydroxide as Cl2 + 2NaOH -> NaCl + NaClO + H2O. Here chlorine (oxidation state 0) is reduced to Cl- (-1) in NaCl and oxidised to ClO- (+1) in sodium hypochlorite, so the same element undergoes both oxidation and reduction.

Exercise - Extended Response Questions (ERQs)

Q(i). Elaborate on the Bronsted-Lowry theory and how it applies to the basicity of ammonia, including the proton transfer reaction.

Ans. The Bronsted-Lowry theory defines an acid as a proton (H+) donor and a base as a proton acceptor. Ammonia is a base because its nitrogen atom carries a lone pair that accepts a proton: NH3 + H+ -> NH4+. The proton-transfer reaction with water is: NH3 + H2O <=> NH4+ + OH-. Here NH3 accepts a proton from water (acting as a base) and water donates it (acting as an acid); the

hydroxide ion produced makes the solution basic. The reverse reaction also occurs: NH4+ donates a proton back, so NH4+ is the conjugate acid of NH3, and OH- is the conjugate base of water. Ammonia can also act as a base towards other acids, for example NH3 + HCl -> NH4Cl.

Q(ii). Provide a comprehensive overview of the catalytic processes used for removing NOx from exhaust gases, highlighting their mechanisms and significance.

Ans. Nitrogen oxides (NOx) from engines are removed by catalytic processes. (1) Three-way catalytic converters (using platinum, palladium and rhodium) simultaneously convert NOx to harmless N2, carbon monoxide to CO2 and unburnt hydrocarbons to CO2 and water. (2) Selective catalytic reduction (SCR): ammonia (from urea) is injected into the exhaust over a vanadia-titania catalyst, reducing NOx to N2 and water (4NO + 4NH3 + O2 -> 4N2 + 6H2O); it is widely used in diesel vehicles and power plants. (3) Lean NOx traps (LNT): NOx is stored on a barium-based trap during lean (oxygen-rich) running and periodically reduced to N2 during short fuel-rich phases. These processes matter because NOx causes smog, acid rain and respiratory disease, and their removal is required by emission laws.

Q(iii). Discuss the relative reactivity of halogens as oxidising agents and explain the trend observed.

Ans. Halogens are strong oxidising agents because they need only one electron to complete their octet and have high electron affinities and electronegativities. Their oxidising strength decreases

down the group: F2 > Cl2 > Br2 > I2. Reasons: atomic size increases down the group, so the incoming electron is farther from the nucleus and less strongly attracted; electron affinity and electronegativity fall (with fluorine slightly anomalous in electron affinity because of repulsion in its small 2p orbitals, yet still the strongest oxidiser overall because its bond energy and hydration energy are very favourable). Consequences: a more reactive halogen displaces a less reactive one from its salts — for example, chlorine displaces bromine and iodine (Cl2 + 2KBr -> 2KCl + Br2), but bromine cannot displace chlorine. The reducing power of halide ions increases in the opposite order: F- < Cl- < Br- < I-.

Q(iv). Discuss the role of chlorine in water purification, including the reactions involved and its importance.

Ans. Chlorine is used in water purification because it kills bacteria, viruses and other microorganisms. When chlorine dissolves in water it reacts: Cl2 + H2O <=> HOCl + HCl. The hypochlorous acid (HOCl) formed is a powerful oxidising agent that destroys the cell walls and enzymes of microbes; it partly dissociates as HOCl <=> H+ + OCl-, and the hypochlorite ion (OCl-) also disinfects and remains in the water as a residual disinfectant, protecting it during storage and distribution. Importance: chlorination prevents waterborne diseases such as cholera and typhoid; in small, controlled doses its health benefits greatly outweigh its risks, which is why municipal plants chlorinate drinking water.

Unit 10: Environmental Chemistry: Atmosphere and Water

Exercise - Multiple Choice Questions (MCQs)

Q(i). The layer of the atmosphere responsible for protecting the Earth from harmful ultraviolet radiation is:

- Troposphere
- ✓ Stratosphere
- Mesosphere
- Thermosphere

Q(ii). The primary cause of water scarcity threatening civilisations is:

- Climate change
- Industrial pollution
- ✓ Unequal distribution
- Marine dumping

Q(iii). The Great Smog of 1952 in London was primarily an example of:

- ✓ Industrial smog
- Photochemical smog
- Natural fog
- Greenhouse effect

Q(iv). The primary cause of acid rain is:

- Carbon dioxide (CO_2)

Methane (CH4)

✓ Sulphur dioxide (SO2)

CFCs

Q(v). The greenhouse gas that contributes the highest percentage to global warming is:

Methane (CH4)

Nitrous oxide (N2O)

✓ Carbon dioxide (CO2)

Fluorinated gases (CFCs)

Q(vi). The reaction contributing to the formation of sulfuric acid in acid rain is:

CO2(g) + H2O(l) -> H2CO3(aq)

2NO(g) + O2(g) -> 2NO2(g)

✓ 2SO2 + O2 -> 2SO3

SO2 + H2 -> H2S

Q(vii). The burning of fossil fuels mainly contributes to:

Depletion of ozone layer

Acid rain formation

✓ Increase in atmospheric CO2 and global warming

Reduction of atmospheric nitrogen

Q(viii). The type of pollution that originates from diffuse sources and is harder to trace back to a specific origin is:

Point source pollution

✓ Nonpoint source pollution

Microbial pollution

Groundwater pollution

Q(ix). The primary purpose of coagulation in water treatment is:

Removal of dissolved salts

Disinfection of water

✓ Clumping of small suspended particles

Filtration of microorganisms

Q(x). An example of nonpoint source water pollution is:

Wastewater discharge from a factory

Oil spill from a tanker

✓ Runoff from agricultural fields

Leakage from a septic tank

Exercise - Restricted Response Questions (RRQs)

Q(i). What are the four major regions of the atmosphere, and what are their altitudes and temperature ranges?

Ans. The four major regions of the atmosphere are: (1) Troposphere — from the surface to about 12 km; temperature decreases with height (about 15 degrees C at the surface to about -56 degrees C at the top); it contains most of the air and almost all weather. (2) Stratosphere — about 12 to 50 km; temperature increases with height (to about 0 degrees C) because the ozone layer absorbs ultraviolet radiation. (3) Mesosphere — about 50 to 85 km; temperature decreases again, to about -90 degrees C, the coldest part. (4) Thermosphere — above about 85 km; temperature rises steeply (above 1000 degrees C) because the thin gases absorb

high-energy solar radiation.

Q(ii). Briefly describe the troposphere and its significance.

Ans. The troposphere is the lowest layer of the atmosphere, extending from the surface to about 12 km. It contains about 75% of the atmosphere's mass and almost all its water vapour. Its significance: nearly all weather (clouds, rain, storms) occurs here; temperature decreases with height, which drives convection and the water cycle; and it is the layer in which we live and breathe, so air pollution in this layer directly affects health.

Q(iii). What health issues are associated with exposure to lead (Pb)?

Ans. Exposure to lead (Pb) causes serious health issues: in children it damages the developing brain and nervous system, lowering intelligence and causing learning and behavioural problems; in adults it raises blood pressure, damages the kidneys, and causes anaemia by interfering with haemoglobin formation; it also harms the reproductive system and, at high levels, causes convulsions, coma and death. Lead accumulates in the body, so even low-level, long-term exposure is dangerous.

Q(iv). Name two renewable energy sources mentioned as strategies for improving air quality.

Ans. Two renewable energy sources mentioned as strategies for improving air quality are solar energy (using sunlight to generate electricity without burning fuel) and wind energy (using wind turbines). Both replace fossil-fuel power stations, cutting emissions of CO2, SO2 and NOx and so reducing smog, acid rain and the

greenhouse effect.

Q(v). What is water scarcity, and what are its main causes?

Ans. Water scarcity is the lack of sufficient available water to meet the needs of a population, region or ecosystem. Its main causes are: unequal distribution of water resources; overuse and wastage, especially in agriculture and industry; pollution, which makes available water unusable; population growth and urbanisation, which raise demand; climate change, which alters rainfall patterns; and poor management and leaky infrastructure, which lose large amounts of water.

Q(vi). What is the significance of potable water, and why is water treatment essential?

Ans. Potable water is water that is safe to drink — free from harmful microorganisms, toxic chemicals and unpleasant taste or odour. Water treatment is essential because raw water from rivers, lakes and groundwater contains germs, suspended solids, dissolved pollutants and chemicals that cause diseases such as cholera, typhoid and dysentery. Treatment makes water safe, protects public health and ensures a reliable supply for drinking, cooking and hygiene.

Q(vii). Briefly explain the process of chlorination in water treatment.

Ans. Chlorination is the final disinfection step in water treatment: chlorine gas (or a chlorine compound) is added to filtered water. It reacts with water to form hypochlorous acid (HOCl) and hypochlorite ions (OCl^-), which are powerful oxidising agents that

kill bacteria, viruses and other pathogens by destroying their cell walls and enzymes. A small residual amount of chlorine is left in the water to keep it safe during storage and distribution.

Q(viii). Name three alternative water treatment methods besides chlorination.

Ans. Three alternative water treatment methods besides chlorination are: (1) Ozonation — passing ozone (O3) through water; ozone is a strong oxidiser that kills microbes and breaks down organic matter without leaving a chemical taste. (2) Ultraviolet (UV) treatment — exposing water to UV light, which damages the DNA of microorganisms and prevents them from reproducing. (3) Boiling and filtration through activated carbon — boiling kills pathogens, while activated carbon adsorbs organic chemicals, odours and colours; reverse osmosis (membrane filtration) is another method that removes dissolved salts and contaminants.

Exercise - Extended Response Questions (ERQs)

Q(i). Explain the greenhouse effect process and its role in maintaining Earth's temperature. How does an enhanced greenhouse effect contribute to global warming?

Ans. The greenhouse effect is the natural warming of the Earth by gases in the lower atmosphere. Sunlight passes through the atmosphere and warms the Earth's surface; the surface radiates infrared (heat) radiation back, and greenhouse gases (CO2, methane, water vapour, nitrous oxide) absorb some of this radiation and re-radiate it in all directions, including back to the

surface — like the glass of a greenhouse. This keeps the Earth's average temperature around 15 degrees C, warm enough for life; without it the Earth would be about -18 degrees C. The enhanced greenhouse effect occurs when human activities (burning fossil fuels, deforestation, industry) add extra greenhouse gases, trapping more heat: global temperatures rise, glaciers melt and sea levels rise, rainfall patterns change, and extreme weather events become more frequent — this is global warming.

Q(ii). Discuss the formation and adverse effects of photochemical smog.

Ans. Photochemical smog forms when nitrogen oxides (NOx) and volatile organic compounds (VOCs) from vehicles and industry react in strong sunlight. Sunlight splits NO2 into NO and atomic oxygen; the oxygen combines with O2 to form ozone (O3), and reactions with VOCs produce peroxyacetyl nitrate (PAN) and aldehydes. Adverse effects: ozone and PAN irritate the eyes, nose and lungs, worsen asthma and bronchitis, and reduce crop yields by damaging plant leaves; the brownish haze reduces visibility; and ozone also damages rubber, paint and other materials. Unlike industrial (London) smog, which is a winter mixture of smoke and fog, photochemical smog is a sunny-weather, oxidising smog, common in cities such as Los Angeles.

Q(iii). Explain the various methods involved in water treatment, including pre-treatment, coagulation, sedimentation, and filtration.

Ans. Water treatment involves several steps. (1) Pre-treatment (screening): large debris, grit and floating matter are removed by

screens and settling. (2) Coagulation: chemicals such as alum are added; they neutralise the charges on tiny suspended particles, which then clump together into larger flocs. (3) Sedimentation: the water is held still so the flocs settle to the bottom as sludge, which is removed. (4) Filtration: the water passes through beds of sand, gravel and activated carbon, which remove remaining particles, odour, taste and some chemicals. After filtration, the water is disinfected (usually by chlorination) and stored for distribution.

Q(iv). Evaluate the ongoing challenges in water conservation in Pakistan, considering factors such as enforcement, wastewater treatment, and infrastructure.

Ans. Pakistan faces serious water-conservation challenges. Enforcement: water laws and regulations exist but are weakly enforced — illegal pumping, water theft and unmetered use are common, and polluters are rarely penalised. Wastewater treatment: most municipal and industrial wastewater is discharged untreated into rivers and canals, contaminating the water that downstream users depend on; few treatment plants exist and many do not work. Infrastructure: canals and pipes are old and leaky, losing large amounts of water; storage capacity is low, so flood water is wasted and dry-season shortages follow; and flood irrigation wastes water compared with drip and sprinkler systems. Solutions include repairing infrastructure, building storage, treating and reusing wastewater, metering and pricing water, and educating farmers and the public about conservation.

Q(v). Elaborate on the human impact on water resources,

specifically examining the effects of agriculture and industrial processes on water quantity, quality, ecosystems, and human communities.

Ans. Human activities affect water resources in several ways. Agriculture: irrigation takes about 90% of Pakistan's freshwater; over-extraction lowers water tables, while fertiliser and pesticide runoff pollutes rivers and groundwater with nitrates and chemicals, causing algal blooms and dead zones. Industry: factories discharge heavy metals, dyes and hot water into rivers, poisoning aquatic life and making water unsafe for drinking and farming; cooling-water use also disturbs river temperatures. Effects on ecosystems: pollution and reduced river flows destroy wetlands and fish habitats. Effects on communities: contaminated and scarce water spreads disease, forces people (especially women and children) to travel far for water, and creates conflicts between users — so sustainable management, treatment and efficient use are essential.

Unit 11: Fundamentals of Organic Chemistry

Exercise - Multiple Choice Questions (MCQs)

Q(i). The property of carbon that allows it to form long chains and cyclic structures is:

- Isomerism
- ✓ Catenation
- Solubility
- Flammability

Q(ii). The name of a group of organic compounds with the same functional group and a gradual change in structure with an increase in the number of carbon atoms is:

- Isomers
- ✓ Homologous series
- Metamers
- Tautomers

Q(iii). In which type of reaction does a nucleophile attack and replace a leaving group?

- Electrophilic addition
- ✓ Nucleophilic substitution
- Free-radical substitution
- Electrophilic substitution

Q(iv). The type of isomers that have the same molecular formula

but different positions of the same functional group is:

- Chain isomers
- ✓ Position isomers
- Functional group isomers
- Metamers

Q(v). The phenomenon where two forms of the same compound differ in the position of protons and electrons is called:

- ✓ Tautomerism
- Metamerism
- Isomerism
- Catenation

Q(vi). The geometry around the carbon atom in methane (CH4) is:

- Trigonal planar
- ✓ Tetrahedral
- Linear
- Bent

Q(vii). The order of reactivity of HX in hydrohalogenation of alkenes is:

- ✓ HI > HBr > HCl
- HCl > HBr > HI
- HI < HBr < HCl
- HBr > HI > HCl

Q(viii). The purpose of a dehydrating agent in the dehydration of alcohols is:

To provide oxygen

✓ To remove water

To increase the temperature

To add hydrogen

Q(ix). The primary source of petrochemicals is:

Natural Gas

Coal

✓ Crude Oil

Naphtha

Q(x). The process of breaking down large hydrocarbons into smaller, more valuable ones is called:

Polymerisation

✓ Cracking

Oxidation

Reforming

Q(xi). The process that involves converting ethylene and propylene into plastics like polyethylene and polypropylene is:

Oxidation

Hydrogenation

✓ Polymerisation

Amination

Q(xii). The petrochemical primarily used to produce synthetic rubber is:

Xylene

Methanol

✓ Butadiene

Toluene

Exercise - Restricted Response Questions (RRQs)

Q(i). What is catenation, and why is catenation a unique property of carbon in organic chemistry?

Ans. Catenation is the ability of an atom to form strong bonds with other atoms of the same element, producing long chains, branches and rings. It is a unique property of carbon in organic chemistry because carbon forms very strong C-C single bonds (bond energy about 346 kJ/mol), has a small size and four valence electrons, so it can bond to four other atoms (carbon or other elements) in stable tetrahedral arrangements. Silicon also catenates, but its Si-Si bonds are much weaker, so its chains are short and unstable; carbon alone builds the millions of stable compounds of organic chemistry.

Q(ii). What is isomerism, and why is it significant in organic chemistry?

Ans. Isomerism is the phenomenon in which two or more compounds have the same molecular formula but different structures, and therefore different properties. It is significant in organic chemistry because it explains the huge diversity of organic compounds: for example, C4H10 exists as butane and 2-methylpropane, which have different boiling points and reactivities; and isomers can have very different biological effects (for example, one optical isomer of a drug may heal while the other harms). Studying isomerism is therefore essential for synthesis,

medicine and industry.

Q(iii). Briefly describe the terms nucleophile and electrophile within the context of organic reactions.

Ans. A nucleophile ('nucleus-loving') is a species with a lone pair of electrons or a negative charge that attacks an electron-deficient (electrophilic) centre and donates its electron pair to form a new bond — examples are OH-, CN-, NH3 and Cl-. An electrophile ('electron-loving') is an electron-deficient species that accepts an electron pair — examples are carbocations, the carbon of a carbonyl group, and H+. In organic reactions, nucleophiles attack electrophiles; for example, in the hydrolysis of an alkyl halide, OH- (nucleophile) attacks the carbon (electrophile) and displaces the halide leaving group.

Q(iv). Discuss the unreactive nature of alkanes and the factors contributing to their stability.

Ans. Alkanes are unreactive because they contain only strong, non-polar C-C and C-H sigma bonds. Factors contributing to their stability: the bonds are very strong and hard to break; the bonds are non-polar, so polar reagents and ions do not attack them; there are no lone pairs, pi bonds or vacant orbitals for reagents to attack; and their carbon atoms are saturated (each bonded to four atoms). Alkanes therefore react only under vigorous conditions — with halogens in sunlight (free-radical substitution) or by combustion at high temperature.

Q(v). What is radical substitution, and how does it occur in alkane

reactions?

Ans. Radical (free-radical) substitution is the replacement of a hydrogen atom of an alkane by another atom or group through free-radical intermediates, typically with halogens in sunlight. It occurs in three stages: (1) Initiation — UV light splits a halogen molecule into two free radicals (Cl2 -> 2Cl.). (2) Propagation — a chlorine radical abstracts a hydrogen from the alkane (Cl. + CH4 -> HCl + CH3.), and the methyl radical reacts with Cl2 to give CH3Cl and a new Cl. radical, continuing the chain. (3) Termination — two radicals combine (for example Cl. + Cl. -> Cl2), ending the chain.

Q(vi). Identify the conditions required for geometric isomerism in hydrocarbons.

Ans. Geometric (cis-trans) isomerism in hydrocarbons requires: (1) restricted rotation about a bond — in alkenes, the pi bond prevents rotation about the C=C double bond, freezing the positions of the groups; and (2) each carbon of the double bond must carry two different groups (if either carbon has two identical groups, cis-trans isomers cannot exist). Example: but-2-ene shows cis and trans forms, but but-1-ene does not, because one carbon of its double bond carries two hydrogen atoms.

Q(vii). Differentiate between petrochemicals and chemicals derived from petrochemicals.

Ans. Petrochemicals are the basic chemicals obtained directly from petroleum and natural gas by refining — for example, ethylene, propylene, benzene and toluene. Chemicals derived from petrochemicals are the products made from those basic chemicals

by further processing — for example, polyethylene and polypropylene (plastics) from ethylene and propylene, polyester from xylene, and fertilisers and explosives from ammonia. In short, petrochemicals are the raw materials, and their derivatives are the finished products built from them.

Q(viii). Briefly explain the importance of ethylene and propylene in the plastics industry.

Ans. Ethylene is the most important feedstock of the plastics industry: it polymerises to polyethylene, used in bags, bottles, pipes and packaging; it is also converted to ethylene glycol (antifreeze) and to vinyl chloride for PVC. Propylene polymerises to polypropylene, used in fibres, carpets, ropes, containers and car parts. Together they are the building blocks of the world's most widely used plastics, so their availability and price shape the whole plastics industry.

Exercise - Extended Response Questions (ERQs)

Q(i). Explain the different types of formulas used to represent chemical compounds, including empirical, molecular, condensed, structural, displayed, and skeletal formulas.

Ans. Empirical formula: the simplest whole-number ratio of atoms in a compound (for example, CH2O for glucose). Molecular formula: the actual number of atoms in one molecule (C6H12O6 for glucose). Condensed formula: a shortened structural formula showing how groups are joined (for example, CH3CH2OH for ethanol). Structural formula: shows every atom and bond (for example, H-C-H bonds drawn for methane). Displayed formula:

shows all bonds explicitly, including every C-H bond. Skeletal formula: carbon atoms are shown as corners and ends of lines, with hydrogens on carbon omitted (for example, a zig-zag line for butane, a hexagon for benzene); heteroatoms and functional groups are written out. The more detailed the formula, the more it reveals about bonding and shape, but the simpler forms are quicker for composition and calculations.

Q(ii). Discuss the concept of tautomers, providing examples and explaining the dynamic equilibrium between keto and enol forms.
Ans. Tautomers are isomers that differ in the position of a proton and a double bond and exist in dynamic equilibrium, interconverting rapidly. The classic example is keto-enol tautomerism: a carbonyl compound (keto form, C=O with an alpha hydrogen) interconverts with its enol form (C=C-OH) by moving the alpha hydrogen to the oxygen. For example, acetaldehyde (CH3CHO, keto) is in equilibrium with vinyl alcohol (CH2=CHOH, enol); the keto form dominates (about 99.99%) because the C=O bond is stronger than C=C. The equilibrium is dynamic: individual molecules constantly switch forms, but the overall ratio stays constant, and the enol form can be trapped by reactions that only the enol undergoes.

Q(iii). Compare and contrast the properties of alkanes and alkenes.
Ans. Alkanes are saturated hydrocarbons (single bonds only, CnH2n+2): they are unreactive (except combustion and free-radical substitution), non-polar, insoluble in water, and have relatively low boiling points that rise with chain length. Alkenes are unsaturated

hydrocarbons (at least one C=C double bond, CnH2n): the pi bond makes them much more reactive — they undergo electrophilic addition (for example, with HBr, Br2 water and H2SO4), can be hydrogenated to alkanes, and polymerise to plastics. Alkenes have slightly higher boiling points than alkanes of similar size, show geometric isomerism, and decolourise bromine water, while alkanes do not. Both are non-polar and burn in air to give CO2 and water.

Q(iv). Elaborate on the concept of geometric isomerism in alkenes, providing relevant examples and explanations.

Ans. Geometric isomerism in alkenes arises because rotation about the C=C double bond is restricted by the pi bond, so the groups on the two carbons are locked in place. If each carbon carries two different groups, two isomers exist: cis, in which the same groups lie on the same side of the double bond, and trans, in which they lie on opposite sides. Example: cis-but-2-ene has both methyl groups on the same side (boiling point 4 degrees C), while trans-but-2-ene has them opposite (boiling point 1 degree C, more stable because the bulky groups are farther apart). Cis isomers are usually more polar and less stable than trans isomers. The conditions are: restricted rotation (a double bond) and two different groups on each double-bond carbon.

Q(v). Discuss the hydrohalogenation of alkenes, emphasising its mechanism and Markovnikov's rule.

Ans. Hydrohalogenation is the addition of HX (HCl, HBr, HI) across the C=C double bond of an alkene, giving an alkyl halide. Mechanism (electrophilic addition): (1) the pi bond attacks the H of

HX, forming a carbocation intermediate and X-; (2) X- attacks the carbocation, giving the product. Markovnikov's rule: when HX adds to an unsymmetrical alkene, the hydrogen adds to the carbon that already has more hydrogens (the less substituted carbon), and the halide adds to the more substituted carbon, because that route forms the more stable carbocation. Example: propene + HBr -> 2-bromopropane (major product), not 1-bromopropane. Reactivity order: HI > HBr > HCl, because the H-X bond weakens down the group.

Q(vi). Explain the process of fractional distillation in the refining of petroleum.

Ans. Fractional distillation separates crude oil into fractions by boiling point. The oil is heated to vapour (about 400 degrees C) and fed into the bottom of a tall fractionating column, which is hot at the bottom and cooler at the top. As the vapour rises, it cools; fractions condense at the level where the temperature equals their boiling range and are drawn off: gases at the top, then petrol (gasoline), naphtha, kerosene, diesel, lubricating oil, and bitumen (residue) at the bottom. Shorter hydrocarbons have lower boiling points and collect higher in the column; longer ones condense lower down. Each fraction is then further processed (for example, by cracking) into fuels and petrochemical feedstocks.

Q(vii). How do fossil fuels form, and what are their properties and uses?

Ans. Fossil fuels form from the remains of dead plants and animals buried under sediment millions of years ago. Under high pressure

and temperature, and without oxygen, the organic matter slowly changes: plant remains in swamps become peat, then coal; marine organisms buried under the sea become crude oil and natural gas. Properties: they are rich in carbon and hydrogen, burn exothermically, and are non-renewable — they take millions of years to form but are used up in decades. Uses: coal for electricity and steel-making; crude oil refined into petrol, diesel, kerosene and petrochemicals; natural gas for heating, electricity and as a chemical feedstock. Their burning releases CO_2 and pollutants, causing global warming and air pollution, which is why renewable alternatives are being developed.

Unit 12: Haloalkanes and Amines

Exercise - Multiple Choice Questions (MCQs)

Q(i). The factor that contributes to steric hindrance in tertiary haloalkanes is:

- High bond polarity
- ✓ Bulky alkyl groups
- Low leaving group ability
- Strong nucleophilicity

Q(ii). The order of reactivity of alkyl halides based on bond energy is:

- RCl > RBr > RI > RF
- ✓ RI > RBr > RCl > RF
- RF > RCl > RBr > RI
- RCl > RF > RBr > RI

Q(iii). The product formed when haloalkanes react with aqueous NaOH is:

- Alkyl cyanides
- Nitroalkanes
- ✓ Alcohols
- Amines

Q(iv). The reaction commonly used to identify haloalkanes is:

- Reaction with aqueous NaOH

Reaction with KCN

Reaction with NH3

✓ Reaction with AgNO3

Q(v). The result of the elimination reaction of bromoethane with sodium hydroxide and heat is:

Formation of an alcohol

✓ Formation of a double bond

Formation of a nitroalkane

Formation of an alkyl cyanide

Q(vi). The primary factor contributing to the basicity of amines is:

Presence of nitrogen

✓ Presence of lone pair on nitrogen

Number of alkyl groups

Molecular weight of amines

Q(vii). The type of amine that is the most reactive and capable of diverse chemical reactions is:

✓ Primary amine

Secondary amine

Tertiary amine

Quaternary ammonium salt

Q(viii). The functional group that characterises all amines is:

-OH

-COOH

✓ -NH2

-CHO

Q(ix). The property that makes aniline soluble in organic solvents is:

Amino group

✓ Benzene ring

Colour

Odour

Q(x). The class of amines that exhibits the least reactivity and nucleophilicity is:

Primary amines

Secondary amines

Tertiary amines

✓ Quaternary amines

Exercise - Restricted Response Questions (RRQs)

Q(i). How does bond polarity influence the reactivity of haloalkanes?

Ans. The carbon-halogen bond is polar because the halogen is more electronegative than carbon, so the carbon becomes partially positive (delta+) and the halogen partially negative (delta-). This polarisation makes the carbon an electrophilic centre that nucleophiles can attack, and the halogen a good leaving group. The more polar the bond, the more susceptible the haloalkane is to nucleophilic attack; however, reactivity also depends on bond strength and steric hindrance, so polarity alone does not decide the order of reactivity.

Q(ii). Write the reaction of the electrophilic addition of an alkene with halogen or hydrogen halide.

Ans. Electrophilic addition of a halogen to an alkene: C2H4 + Br2 -> CH2Br-CH2Br (1,2-dibromoethane). The pi bond attacks a bromine molecule, forming a cyclic bromonium ion and Br-; the bromide ion then attacks from the back, giving the dibromide. Electrophilic addition of a hydrogen halide: C2H4 + HBr -> CH3CH2Br (bromoethane); the pi bond takes H+ to form a carbocation, which is then attacked by Br-. With unsymmetrical alkenes, Markovnikov's rule applies.

Q(iii). What is the major product obtained when bromoethane reacts with aqueous silver nitrate in ethanol?

Ans. The major organic product is ethoxyethane (diethyl ether, C2H5OC2H5), formed when ethanol acts as the nucleophile. Silver ions pull the bromide off as a pale-yellow precipitate of AgBr, leaving a carbocation that ethanol attacks; loss of a proton gives the ether. (A small amount of ethene is also formed by elimination, and the AgBr precipitate is the visible test for the halide.)

Q(iv). How are amines classified based on the number of alkyl groups bonded to the nitrogen atom?

Ans. Amines are classified by the number of alkyl (or aryl) groups bonded to the nitrogen atom: primary (1 degree) amines have one alkyl group (R-NH2, for example methylamine); secondary (2 degree) amines have two (R2NH, for example dimethylamine); tertiary (3 degree) amines have three (R3N, for example trimethylamine). A quaternary ammonium salt has four groups

(R4N+), carries a positive charge and is not an amine but an ammonium salt.

Q(v). Why are amines considered basic in nature?

Ans. Amines are basic because the nitrogen atom carries a lone pair of electrons that it can donate to a proton (H+) or to another electron-deficient species. For example, NH3 + H+ -> NH4+, and CH3NH2 + HCl -> CH3NH3+Cl-. The availability of this lone pair makes amines behave as Bronsted-Lowry bases and as nucleophiles; alkyl groups increase basicity by pushing electrons towards nitrogen, while the delocalisation of the lone pair into a benzene ring (as in aniline) decreases it.

Q(vi). How are amines regenerated from the salts obtained during alkylation?

Ans. During alkylation, an amine reacts with an alkyl halide and forms its ammonium salt (for example, RNH2 + RX -> RNH3+X-). The free amine is regenerated by treating the salt with a base (such as aqueous NaOH or ammonia): RNH3+X- + OH- -> RNH2 + H2O + X-. The base removes the extra proton from the ammonium ion, restoring the lone pair on nitrogen and giving back the free amine.

Q(vii). How are nitro compounds converted to primary amines?

Ans. Nitro compounds are converted to primary amines by reduction. Common methods: (1) Tin and concentrated hydrochloric acid, followed by alkali (Sn/HCl, then NaOH) — for example, nitrobenzene -> phenylamine (aniline). (2) Catalytic hydrogenation with hydrogen over a nickel or platinum catalyst (H2/Ni or H2/Pt).

(3) Iron and hydrochloric acid (Fe/HCl) or lithium aluminium hydride (LiAlH4). In each case the -NO2 group gains hydrogen (or loses oxygen) stepwise through nitroso and hydroxylamine intermediates to give -NH2.

Exercise - Extended Response Questions (ERQs)

Q(i). Provide a detailed explanation of the nucleophilic substitution reactions of haloalkanes, focusing on the reactions with aqueous NaOH, KCN, and NH3.

Ans. Nucleophilic substitution in haloalkanes replaces the halogen by a nucleophile. With aqueous NaOH, OH- attacks the electrophilic carbon and displaces the halide, giving an alcohol (R-X + OH- -> R-OH + X-). With KCN, CN- attacks to give an alkyl cyanide (nitrile, R-CN), which is useful because it extends the carbon chain by one atom. With NH3, ammonia attacks to give a primary amine (R-X + NH3 -> R-NH2 + HX), though further alkylation can give secondary and tertiary amines. Mechanisms: SN2 (one step, backside attack, favoured by primary haloalkanes and strong nucleophiles) and SN1 (two steps via a carbocation, favoured by tertiary haloalkanes and polar solvents). Reactivity follows bond strength and leaving-group ability: RI > RBr > RCl > RF, and tertiary > secondary > primary for SN1, while the reverse holds for SN2 because of steric hindrance.

Q(ii). Compare and contrast the bond energies of C-F, C-Cl, C-Br, and C-I bonds, and explain their impact on the reactivity of alkyl halides.

Ans. Bond energies: C-F is about 467 kJ/mol, C-Cl about 346 kJ/mol, C-Br about 290 kJ/mol and C-I about 228 kJ/mol. The strength

decreases down the group because the halogen atoms get larger, so the overlap between carbon's orbital and the halogen's orbital becomes poorer and the bond longer and weaker. Impact on reactivity: a weaker bond breaks more easily, so the halogen leaves more readily, giving the reactivity order RI > RBr > RCl > RF in nucleophilic substitution. This is why alkyl iodides react fastest and alkyl fluorides are almost inert, despite the C-F bond being the most polar — bond strength, not polarity alone, controls the rate.

Q(iii). Discuss the process of substitution of an alcohol to form a haloalkane, including the reagents and conditions involved.
Ans. An alcohol is converted to a haloalkane by replacing its -OH group with a halogen. Reagents and conditions: (1) With HCl in the presence of anhydrous ZnCl2 (Lucas reagent) — tertiary alcohols react at once, secondary on warming, primary only on heating — giving the chloroalkane. (2) With PCl5 or PCl3, or with SOCl2 (thionyl chloride, preferred because its by-products SO2 and HCl are gases, leaving a pure product): R-OH + SOCl2 -> R-Cl + SO2 + HCl. (3) With HBr or a mixture of KBr and concentrated H2SO4 for bromoalkanes, and with HI or KI/H3PO4 for iodoalkanes. The mechanism involves protonating the -OH to make water, a good leaving group, and then substitution by the halide.

Q(iv). Compare and contrast the reactivity, nucleophilicity, and basicity of primary, secondary, and tertiary amines. Provide examples to support your answer.
Ans. Reactivity and nucleophilicity: primary amines (RNH2) are the most reactive and nucleophilic — their nitrogen lone pair is freely

available and the molecule is unhindered, so they readily attack electrophiles and form amides, imines and diazonium salts (from aromatic primary amines). Secondary amines (R2NH) are also good nucleophiles, but the extra alkyl group adds steric hindrance. Tertiary amines (R3N) are the most basic in the gas phase (three electron-donating alkyl groups), but in water their basicity and nucleophilicity are reduced by steric hindrance and poorer solvation of the resulting ammonium ion; the observed aqueous basicity order is usually secondary > primary > tertiary. Basicity: alkyl groups increase electron density on nitrogen (raising basicity), while aryl groups decrease it by delocalising the lone pair into the ring — aniline is much less basic than aliphatic amines. Examples: methylamine (strong base, pKb about 3.4) vs aniline (weak base, pKb about 9.4).

Q(v). Provide information on the properties, reactivity, and applications of phenylamine (aniline). Discuss its use in organic syntheses and potential hazards.

Ans. Phenylamine (aniline, C6H5NH2) is a colourless oily liquid with a characteristic smell, slightly soluble in water but soluble in organic solvents (due to its benzene ring), boiling point 184 degrees C. Reactivity: it is a weak base (its lone pair is delocalised into the ring) and a nucleophile; it undergoes electrophilic substitution on the ring (directing ortho/para), acetylation to acetanilide, and diazotisation with NaNO2/HCl at 0-5 degrees C to give benzene diazonium chloride, the key to azo dyes. Applications: manufacture of dyes, pharmaceuticals (for example paracetamol and sulpha drugs), polymers (polyurethane, rubber chemicals) and

photographic chemicals. Hazards: aniline is toxic — it is absorbed through the skin and lungs, converts haemoglobin to methaemoglobin (causing blue lips and breathlessness), and is a suspected carcinogen, so it must be handled with gloves and ventilation.

Q(vi). Explain the formation of azo compounds through coupling reactions. Discuss the general formula of azo compounds and their colour variations.

Ans. Azo compounds contain the azo group (-N=N-) linking two groups, with the general formula R-N=N-R' (for aromatic azo compounds, Ar-N=N-Ar'). They form by a coupling reaction: a diazonium salt (made by diazotising a primary aromatic amine with NaNO2 and HCl at 0-5 degrees C) reacts with an electron-rich aromatic compound (a phenol or an aromatic amine) in a mildly alkaline or acidic medium, for example benzene diazonium chloride + phenol -> p-hydroxyazobenzene (orange). Colour variations: the colour depends on the groups attached and on the length of the conjugated system — simple azo compounds are yellow to orange, while larger conjugated systems and auxochromes (-OH, -NH2, -SO3H) shift the colour to red, violet, blue or black. Azo dyes are the largest class of synthetic dyes, used for textiles, leather, paper, food colours and inks; azo pigments (insoluble) are used in paints and plastics.

Unit 13: Alcohols and Carbonyl Compounds

Exercise - Multiple Choice Questions (MCQs)

Q(i). The organic compounds containing a hydroxyl functional group are:

✓ Alcohols
Aldehydes
Ketones
Alkyl halides

Q(ii). On the basis of the number of hydroxyl groups alcohols are classified into:

Primary and secondary alcohols
✓ Monohydric and polyhydric alcohols
Saturated and unsaturated alcohols
Aromatic and aliphatic alcohols

Q(iii). The alcohol produced by the reduction of an aldehyde is:

✓ Primary alcohol
Secondary alcohol
Tertiary alcohol
Polyhydric alcohol

Q(iv). The product of the oxidation of a primary alcohol is:

Ketone
✓ Aldehyde

Carboxylic acid

Ester

Q(v). The method used to distinguish primary, secondary, and tertiary alcohols is:

Iodoform test

✓ Oxidation test

Lucas test

Dehydration test

Q(vi). Aldehydes are detected in the 2,4-DNPH test by the formation of a precipitate with a specific colour:

Red precipitate

Grey-black precipitate

✓ Orange-yellow precipitate

Yellow precipitate

Q(vii). The reagent commonly used in the reduction of aldehydes and ketones to alcohols is:

✓ $NaBH_4$

HCN

$K_2Cr_2O_7$

H_2SO_4

Q(viii). The test used to detect the acetyl group in aldehydes and ketones is:

Fehling's test

Tollen's test

✓ Iodoform test

2,4-DNPH test

Q(ix). The metal hydride commonly used as a reducing agent for aldehydes and ketones is:

✓ NaBH4

LiAlH4

K2Cr2O7

HCN

Q(x). The product of the hydrolysis of esters with dilute alkali is:

Carboxylic acid and alcohol

Alkane and water

✓ Carboxylate salt and alcohol

Alkene and carbon dioxide

Exercise - Restricted Response Questions (RRQs)

Q(i). Name two types of hydroxy compounds and provide examples of each.

Ans. Two types of hydroxy compounds are: (1) Alcohols — compounds in which the -OH group is attached to a saturated carbon atom; examples are methanol (CH3OH) and ethanol (C2H5OH). (2) Phenols — compounds in which the -OH group is attached directly to a benzene ring; the example is phenol (C6H5OH). (Carboxylic acids also contain an -OH group, as part of the -COOH group; an example is acetic acid, CH3COOH.)

Q(ii). Differentiate between primary, secondary, and tertiary

alcohols.

Ans. Primary alcohols have the -OH group on a carbon attached to only one other carbon (for example ethanol, CH3CH2OH); they oxidise first to aldehydes and then to carboxylic acids. Secondary alcohols have the -OH group on a carbon attached to two other carbons (for example propan-2-ol); they oxidise to ketones and no further. Tertiary alcohols have the -OH group on a carbon attached to three other carbons (for example 2-methylpropan-2-ol); they resist oxidation because the carbon has no hydrogen to lose. Acidity falls in the order primary > secondary > tertiary, and reactivity towards Lucas reagent rises in the opposite order.

Q(iii). What happens during the combustion of organic hydroxy compounds?

Ans. During the combustion of an organic hydroxy compound (for example ethanol), the compound burns completely in excess oxygen to give carbon dioxide and water, releasing heat: C2H5OH + 3O2 -> 2CO2 + 3H2O. In a limited supply of oxygen, incomplete combustion occurs, giving carbon monoxide and/or carbon (soot) along with water. Combustion is the basis of the use of alcohols such as ethanol as fuels.

Q(iv). Give an example of a monohydric alcohol.

Ans. An example of a monohydric alcohol is ethanol (C2H5OH) — it contains only one hydroxyl (-OH) group per molecule. Methanol (CH3OH) is another example. (By contrast, glycol, CH2OH-CH2OH, is dihydric, and glycerol is trihydric.)

Q(v). Name the reducing agent commonly used for the reduction of aldehydes and ketones.

Ans. The reducing agent commonly used for the reduction of aldehydes and ketones is sodium borohydride (NaBH4) in methanol or water. It reduces aldehydes to primary alcohols and ketones to secondary alcohols, without attacking carbon-carbon double bonds. (Lithium aluminium hydride, LiAlH4, is a stronger reducing agent used for the same purpose, and also reduces carboxylic acids and esters.)

Q(vi). Provide the mechanism for the addition of hydrogen cyanide to a carbonyl compound.

Ans. Mechanism of HCN addition to a carbonyl compound (for example ethanal): (1) The cyanide ion (CN-), a nucleophile, attacks the electrophilic carbonyl carbon, because oxygen pulls the pi electrons towards itself, leaving carbon partially positive. The pi bond breaks and the electron pair moves to oxygen, giving an alkoxide intermediate with the CN group attached to carbon. (2) The alkoxide ion takes a proton from HCN (or from water/acid in the work-up), giving the cyanohydrin product (for example, CH3CH(OH)CN from ethanal) and regenerating CN-. The reaction is base-catalysed and is an example of nucleophilic addition.

Q(vii). Why is it important to use freshly prepared Tollen's reagent in the Silver Mirror test?

Ans. Tollen's reagent (ammoniacal silver nitrate, [Ag(NH3)2]+OH-) must be freshly prepared because, on standing, it can form silver nitride (Ag3N), a highly explosive compound, especially if the

solution is allowed to dry or is stored. Fresh reagent is also more reliable: old reagent may give a silver mirror even with ketones or fail with aldehydes, ruining the test. For safety and accuracy, it is made just before use and any residue is destroyed with dilute acid immediately afterwards.

Q(viii). List three methods for the preparation of carboxylic acids.
Ans. Three methods for the preparation of carboxylic acids are: (1) Oxidation of primary alcohols (or aldehydes) with acidified potassium dichromate or potassium permanganate under reflux — for example, ethanol -> ethanoic acid. (2) Hydrolysis of nitriles (alkyl cyanides) with dilute acid or alkali, followed by acidification — for example, CH3CN -> CH3COOH. (3) Hydrolysis of esters with dilute acid or alkali — for example, ethyl ethanoate -> ethanoic acid + ethanol. (A fourth method is the reaction of a Grignard reagent with carbon dioxide, followed by acid work-up.)

Q(ix). Define esterification and mention the catalyst typically used in this reaction.
Ans. Esterification is the reaction of a carboxylic acid with an alcohol to form an ester and water: R-COOH + R'-OH <=> R-COOR' + H2O. The catalyst typically used is concentrated sulphuric acid, which also acts as a dehydrating agent, removing the water and driving the equilibrium towards the ester. The reaction is reversible and is usually carried out under reflux; for example, ethanoic acid + ethanol -> ethyl ethanoate (a sweet-smelling liquid) + water.

Exercise - Extended Response Questions (ERQs)

Q(i). Discuss the oxidation of alcohols and how it helps differentiate between primary, secondary, and tertiary alcohols.

Ans. Oxidation of alcohols with acidified potassium dichromate (orange) or potassium permanganate (purple) distinguishes the three classes. A primary alcohol oxidises first to an aldehyde (the dichromate turns green) and, under reflux, further to a carboxylic acid — for example, ethanol -> ethanal -> ethanoic acid. A secondary alcohol oxidises to a ketone and stops there, because the carbonyl carbon has no hydrogen left to remove — for example, propan-2-ol -> propanone. A tertiary alcohol does not oxidise at all (the dichromate stays orange), because its carbon carries no hydrogen atom. So the oxidation test (colour change and product) identifies the class: aldehyde/acid = primary, ketone = secondary, no reaction = tertiary.

Q(ii). Describe the dehydration of alcohols, including the factors influencing the reaction and the order of ease.

Ans. Dehydration of alcohols removes water from the molecule (using concentrated sulphuric acid or phosphoric acid, with heat) to give an alkene: for example, ethanol -> ethene + water. Factors influencing the reaction: temperature (higher temperature favours elimination over substitution); the acid catalyst (concentrated H2SO4, about 170 degrees C for ethanol); and the structure of the alcohol. Order of ease: tertiary alcohols dehydrate most easily, then secondary, then primary (3 degree > 2 degree > 1 degree), because the reaction goes through a carbocation intermediate, and tertiary carbocations are the most stable. The major product follows Saytzeff's rule: the hydrogen is removed from the carbon with

fewer hydrogens, giving the more substituted (more stable) alkene.

Q(iii). Describe the iodoform test and how it distinguishes between methanol and ethanol.

Ans. The iodoform test detects the CH3CO- (acetyl) group or the CH3CH(OH)- group: the compound is warmed with iodine and sodium hydroxide, and a yellow precipitate of iodoform (CHI3) with an antiseptic smell appears if the group is present. Ethanol (CH3CH2OH) contains the CH3CH(OH)- group, so it is first oxidised to ethanal (CH3CHO) and then gives a positive test (yellow precipitate). Methanol (CH3OH) has no methyl group attached to the carbon bearing the -OH, so it cannot form the acetyl intermediate and gives a negative test (no precipitate). The test therefore distinguishes ethanol (positive) from methanol (negative).

Q(iv). Detail the preparation methods of alcohols, including electrophilic addition of steam to an alkene and reduction of aldehydes.

Ans. Preparation methods of alcohols: (1) Electrophilic addition of steam to an alkene — steam and an alkene are passed over a phosphoric acid catalyst at about 300 degrees C and 60-70 atmospheres (for example, ethene + H2O -> ethanol). This is the industrial method for ethanol. (2) Reduction of aldehydes — aldehydes are reduced with NaBH4 or LiAlH4 (or H2/Ni) to primary alcohols (for example, ethanal -> ethanol); reduction of ketones gives secondary alcohols. (3) Hydrolysis of alkyl halides with aqueous NaOH or KOH (R-X + OH- -> R-OH + X-). (4) Hydration of alkenes with dilute sulphuric acid followed by hydrolysis

(Markovnikov addition). (5) Fermentation of sugars by yeast for ethanol (in dilute solution).

Q(v). Explain the principles behind the three tests used to identify carbonyl compounds: 2,4-DNPH, Fehling, and Tollen's.
Ans. (1) 2,4-DNPH test: 2,4-dinitrophenylhydrazine (Brady's reagent) reacts with the carbonyl group of both aldehydes and ketones to give an orange or yellow precipitate (a 2,4-dinitrophenylhydrazone); a precipitate confirms a carbonyl compound, but the test does not distinguish aldehydes from ketones. (2) Fehling's test: Fehling's solution (an alkaline copper(II) complex, blue) is reduced by aldehydes to a brick-red precipitate of copper(I) oxide on warming, while ketones do not react — so it identifies aldehydes (and reducing sugars). (3) Tollen's test (silver mirror test): ammoniacal silver nitrate is reduced by aldehydes to metallic silver, which coats the tube as a mirror, while ketones do not react (except alpha-hydroxy ketones). Together, 2,4-DNPH proves a carbonyl group, and Fehling's and Tollen's tests prove it is an aldehyde rather than a ketone.

Q(vi). Provide a step-by-step explanation of the preparation of carboxylic acids through the oxidation of primary alcohols.
Ans. Step by step: (1) A primary alcohol (for example ethanol, CH_3CH_2OH) is heated under reflux with an oxidising agent, acidified potassium dichromate ($K_2Cr_2O_7/H_2SO_4$) or potassium permanganate. (2) The oxidiser removes two hydrogen atoms from the -CH_2OH group, first forming the aldehyde (ethanal, CH_3CHO); the dichromate changes from orange to green as it is reduced. (3)

Because the mixture is under reflux, the aldehyde stays in contact with the oxidiser and is oxidised further: the aldehyde's C-H bond is replaced by C-OH, giving the carboxylic acid (ethanoic acid, CH3COOH). (4) If the aldehyde is wanted instead, it is removed by distillation as soon as it forms (its boiling point is lower), so it cannot be oxidised further. The overall change is: R-CH2OH --[O]--> R-CHO --[O]--> R-COOH.

Unit 14: Organic Synthesis

Exercise - Multiple Choice Questions (MCQs)

Q(i). The primary goal of organic synthesis is:

- Isolation of natural products
- ✓ Creating complex molecules
- Extraction of minerals
- Studying physical properties

Q(ii). In retrosynthetic analysis, the first step is:

- Selection of solvents and catalysts
- Identification of protecting groups
- ✓ Identification of the target molecule
- Choice of purification method

Q(iii). The functional group that is easily oxidised to carboxylic acids is:

- Alkene
- Alkyne
- ✓ Aldehyde
- Alcohol

Q(iv). The functional group that has a basic nature and participates in nucleophilic substitution is:

- Alkane
- Alcohol

✓ Amine

Aldehyde

Q(v). The reduction of p-nitrophenol to p-aminophenol is an example of:

Functional group protection

✓ Functional group interconversion

Bond breaking

Retrosynthetic tree construction

Q(vi). In synthetic route analysis, the method used to confirm the structure of intermediate products is:

Mass spectrometry

Infrared spectroscopy

Nuclear magnetic resonance

✓ All of the above

Q(vii). The type of reaction that involves the transfer of electrons is:

Nucleophilic substitution

Hydrogenation

✓ Oxidation-reduction

Esterification

Q(viii). The major organic product formed in the nucleophilic substitution of ethyl bromide with ammonia is:

Ethene

Ethanol

✓ Ethylamine

Acetamide

Q(ix). The functional group that undergoes esterification to form esters is:

Alkane

Alkene

✓ Alcohol

Aldehyde

Q(x). The primary goal of retro-synthetic analysis in organic chemistry is to:

Determine the pH of a solution

Analyse combustion reactions

✓ Break down complex molecules into simpler starting materials

Measure functional group solubility

Exercise - Restricted Response Questions (RRQs)

Q(i). Describe the key feature of oxidation-reduction reactions in functional group interconversion.

Ans. The key feature of oxidation-reduction (redox) reactions in functional group interconversion is the transfer of electrons, seen as a change in the oxidation state of carbon: oxidation (loss of electrons, or gain of oxygen/loss of hydrogen) raises the oxidation state, while reduction (gain of electrons, or gain of hydrogen/loss of oxygen) lowers it. For example, a primary alcohol (-2) oxidises to an aldehyde (0) and then to a carboxylic acid (+2), and the reverse reductions lower the state again. This lets chemists move step by step between functional groups in a synthesis.

Q(ii). How do functional groups determine the properties and reactivity of organic molecules?

Ans. Functional groups determine the properties and reactivity of organic molecules because chemical reactions happen at the functional group, while the hydrocarbon skeleton mostly stays unchanged. For example, the -OH group makes alcohols polar, able to hydrogen-bond (so higher boiling points and water solubility) and able to be oxidised or dehydrated; the C=C double bond makes alkenes undergo addition reactions; and the -NH2 group makes amines basic and nucleophilic. Two compounds with the same skeleton but different functional groups therefore behave completely differently.

Q(iii). Why is selecting the proper reaction for the first step crucial in multi-step synthetic routes?

Ans. Selecting the proper reaction for the first step is crucial because every later step depends on it: the first step sets the carbon skeleton and the functional groups available for the rest of the route. A wrong first step can introduce a group that interferes with later reactions, produce the wrong regio- or stereoisomer, or lead to a dead end that forces the whole route to be redesigned. It also affects the overall yield, since the yield of a multi-step synthesis is the product of the yields of all steps — a poor first step lowers the final yield and wastes starting material.

Q(iv). What factors should be considered when optimising reaction conditions in multi-step synthesis?

Ans. Factors to consider when optimising reaction conditions in multi-step synthesis are: temperature (high enough for a reasonable rate, low enough to avoid side reactions and decomposition); choice of solvent (polar protic vs aprotic, boiling point, ability to dissolve reactants); catalyst (type and amount); concentration and stoichiometry (avoiding excess that causes side products); pH (especially for acid- or base-sensitive groups); reaction time; and the need for protecting groups to shield reactive groups that must survive a step. Each step must also give a product pure enough (by work-up and purification) for the next step.

Exercise - Extended Response Questions (ERQs)

Q(i). Explain the process of organic synthesis and its significance in the manufacturing of complex organic compounds.

Ans. Organic synthesis is the planned construction of complex organic molecules from simpler, commercially available starting materials through a sequence of chemical reactions. It is significant because: it provides medicines (for example paracetamol, aspirin, antibiotics) that cannot be obtained in sufficient quantity from nature; it creates new materials (plastics, fibres, dyes, adhesives) with designed properties; it allows the total synthesis of natural products (for example vitamins, hormones) for study and use; and it supports agriculture (pesticides, fertilisers) and industry (solvents, fuels). Without synthesis, modern medicine, materials and agriculture would be impossible.

Q(ii). Define functional group interconversion (FGI) and discuss its

importance in organic synthesis. Provide examples of chemical reactions involved in FGI and their applications.

Ans. Functional group interconversion (FGI) is the conversion of one functional group into another within a molecule, without changing the carbon skeleton. It is important in synthesis because the group needed for the next step is often not the group present: FGI prepares the molecule for the key bond-forming step. Examples: oxidation of a primary alcohol to an aldehyde and then to a carboxylic acid; reduction of a nitro group to an amine (p-nitrophenol -> p-aminophenol in the paracetamol route); conversion of an alcohol to an alkyl halide with SOCl2, so that it can undergo nucleophilic substitution; and hydrolysis of an ester to a carboxylic acid and an alcohol. Applications: drug manufacture (paracetamol from phenol via nitration, reduction and acetylation), dyes (conversion of amines to diazonium salts and azo dyes) and polymer production.

Q(iii). Provide a step-by-step guide for devising multi-step synthetic routes in organic chemistry.

Ans. A step-by-step guide to devising a multi-step synthetic route: (1) Identify the target molecule and its functional groups. (2) Work backwards (retrosynthesis): break strategic bonds and convert functional groups (FGI) to reach simpler, available starting materials, building a retrosynthetic tree. (3) Choose the forward route: write the reactions in the forward direction, checking that each step is feasible and that reagents do not attack other groups. (4) Introduce protecting groups where a reactive group must survive a step, and remove them afterwards. (5) Optimise

conditions (temperature, solvent, catalyst, stoichiometry) for each step. (6) Plan purification and work-up after each step, and confirm each intermediate's structure (by IR, NMR, mass spectrometry, melting point). (7) Evaluate the route: fewest steps, highest overall yield, cheapest and safest reagents, and least waste. Example: paracetamol is made from phenol by nitration to p-nitrophenol, reduction to p-aminophenol, and acetylation with acetic anhydride.